Group (Mis)Perception and Strategic Behavior in Protests: Evidence from the Women's March*

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Abstract

We investigate how beliefs about different groups influence protest participation. Using a field experiment during the Women's March in Chile, we randomly provide undergraduate students with information on the participation of groups defined by their family income. With this experiment, we characterize: (i) heterogeneity in perceptions about other groups' participation, and (ii) heterogeneity in strategic incentives with respect to the in-group and the out-group. Students are randomly assigned to six treatments, receiving information about participation intentions from their in-group, out-group, the entire sample, or a combination of these. Our results suggest that participation of higher-income students is highly underestimated, whereas participation of the lower-income group seems to be predicted more accurately. Both groups display strategic complementarity. However, they react to different reference groups: the higher-income group shows in-group and out-group complementarity, while the lower-income group displays only out-group complementarity. We explore potential mechanisms driving these findings.

JEL Codes: J16, D74, D83

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1 Introduction

Protests have become a prevalent phenomenon worldwide. However, there is still no consensus in the literature on the strategic incentives people face when deciding whether to participate in protests and demonstrations. The most prevalent view in political science and economics postulates strategic complementarity, based on the ideas of power in numbers (DeNardo, 2014), coordination (Shadmehr, 2018), or social image concerns (Enikolopov et al., 2020; Gonzalez, 2020; Bursztyn et al., 2021). However, recent empirical evidence challenges this conception, opening the possibility to strategic substitutability (Cantoni et al., 2019; Shadmehr, 2018).

In this paper, we study how beliefs about the behavior of different groups of potential protesters affect agents' decisions to participate in protests. In particular, we explore if strategic incentives can differ depending on what the reference group is. Do people present different strategic incentives with respect to people sharing the same characteristics compared to people more distant from themselves? We first study whether agents can behave as complements with some groups but substitutes with others, and then explore the mechanisms driving these differences.

To answer these questions, we implement a field experiment with undergraduate students from two Chilean universities in the context of the Women's March. We separate students into two groups according to their self-reported family income—low vs. high—which we use to deliver information that can potentially affect their participation decision. Agents are randomly assigned to one of six treatments, with information about the intention to participate in the demonstrations reported by their in-group, the out-group, the entire sample, or a combination of these. With these treatments, we are able to characterize agents' perceptions about other groups' participation, as well as the strategic incentives they face with respect to these groups.

The Women's March in Chile takes place every year on March 8th, to commemorate International Women's Day. This has become one of the country's largest protests held frequently, making it a great candidate for a field intervention.[1] The demonstration usually gathers people who identify as women from different ages, socioeconomic backgrounds, and ideologies. It is known for its peaceful nature, which helps as well in terms of interventions as it imposes very low risks on participants.[2]

[1]Historic participation usually ranges around 300.000 participants, but in the year 2020, participation reached an unprecedented participation of more than a million participants. See Table D-1 for more detail.

[2]In the survey, participants reported an average level of perception of violence of 1.80, where answers ranged

Our main results show heterogeneity in perceptions about groups' participation and heterogeneity in strategic incentives with respect to the in-group and the out-group.

First, we show that participation of the high-income group of students is highly underestimated on average by all students, whereas participation of the low-income group is more accurately predicted. Additionally, both income groups exhibit more precise estimations of their in-group participation. The latter result is consistent with the findings of Bursztyn and Yang (2022), who show that agents' misperceptions about others are higher for out-group than in-group perception. However, in contrast to their result, we find that both groups underestimate the participation of the higher-income group, and this underestimation is consistent across ideologies, ages, and areas of study.

We conjecture that this might be the result of rooted stereotypes that associate upper-class groups in society with more conservative values, less pro-social behavior, and lower political participation. To explore these ideas, we implemented a follow-up survey to dig into agents' beliefs and perceptions about specific income groups. We find that misperceptions are strongly correlated with students' social environment: increased interaction with individuals from the high-income group enhances the accuracy of students' beliefs about them. This holds true for both low- and high-income students, and it highlights the role of intergroup contact and segregation in perpetuating stereotypes (Allport, 1954; Enos and Celaya, 2018).

Our second set of results refers to heterogeneity in strategic incentives to participate. We use the experiment treatments as exogenous shocks to students' beliefs about the participation of higher and lower income groups in the march, to then estimate the causal effect of their beliefs about the in- and out-group participation on their own likelihood to participate. We find that even though agents behave as complements, people from different socioeconomic backgrounds indeed behave strategically differently. While the higher income group demonstrates strategic complementarities with both the in- and out-group, the lower income group only behaves as complements with the out-group. Furthermore, the effect of the out-group expected participation on a student's own participation for the lower income group is more than double the effect of the in-group expected participation, once again reinforcing the importance of heterogeneous strategic incentives and this present study.

An alternative interpretation of the results presented above would be that a higher turnout of higher-income students has different effects overall than a higher turnout of lower-income

from 1=Peaceful to 5=Violent.

students. While a higher turnout by the high-income group increases the expected participation of both the higher and lower-income groups, a higher turnout by the lower-income group appears to be only relevant for the decision of high-income students.

We then study potential drivers of participation to better understand what drives the results. Both past participation and beliefs about groups' participation are found to be significant economically and statistically. While past participation has been widely mentioned by the related literature as an important determinant of current participation, little has been said about beliefs about specific groups' attendance. We find that an increase of 1 percentage point (pp) in the belief about the higher and lower income group turnout increases the probability of attending by 10.6 and 8.87 pp respectively, while global turnout beliefs have no significant effect.

This paper contributes to a wide but still growing empirical literature on the strategic incentives to participate in protests. The work closer to ours is the paper by Cantoni et al. (2019). By implementing a field experiment in Hong Kong, the authors find evidence of strategic substitutes. In contrast to our experiment, the authors do not focus on specific group incentives but only on global incentives. In another related paper, Hager et al. (2022) study the decision to protest when there are threats of counter-protests in Germany. Related to our work, they find heterogeneous strategic incentives on the left and right side of the spectrum: supporters of the right movement behave as substitutes, but those on the left behave as complements (see also Bursztyn et al., 2021 and Hager et al., 2023 for related experiments).

Our results are at odds with the predictions in Shadmehr (2018). His model predicts that when goals are modest (as is the case in the Hong Kong protests), strategic substitutability will dominate, whereas with more relevant goals, one should observe complementarity. In our framework, despite the goal being modest, we only find evidence of complementarity. One possible explanation for this might be that Women's Marches are often perceived as an end in themselves rather than a means to achieve an instrumental outcome. When protests are primarily expressive, it is more natural to observe complementarities.

Finally, our work contributes to the literature on misperceptions about others' behavior (see Bursztyn and Yang, 2022 and the references therein). Our results provide evidence of groups misperception that seems to be consistent with stereotyping of the high-income group (Bordalo et al., 2016). But, interestingly, in our setting this stereotype is prevalent not only with respect to out-group perceptions but also to the in-group. Our findings are also consistent with the evidence of the effect of segregation and the lack of intergroup contact over groups' stereotypes

(Allport, 1954; Enos, 2017) and political outcomes (Ananat and Washington, 2009; Corvalan and Vargas, 2015).

The rest of the paper is organized as follows. In Section 2, we review the theoretical background and conjecture on the diverse mechanisms that may drive our results. In Section 3, we provide an in-depth description of the experimental design. The core empirical analysis, estimations, and the study of mechanisms can be found in Section 4. We discuss our main results and limitations in Section 5.

2 Theoretical Background

There are several theories explaining why people participate in protests. On the one hand, theories focused on the expressive payoffs from participating (Gurr, 1970) see protests as a process of releasing grievances, in which citizens do not necessarily take strategic concerns into account. On the other hand, the approach developed initially by Tilly (1978) proposes that when deciding whether to participate, citizens do take into account the benefits, costs, and possible outcomes. This paper is based on the latter strand of theories, as we focus on strategic participation, i.e., how people's incentives to participate change when others join.

Strategic incentives in protests have been generally studied, taking protests as a unified entity, where potential protesters, despite their heterogeneity, are assumed to respond to changes in the participation of the entire pool of potential participants. However, in many contexts protests involve diverse groups of individuals and then, the decision to participate depends on how these different groups are perceived. Heterogeneous incentives concerning different groups can depend on many factors, which can be specific to the strategic environment or the chosen groups. For instance, it might be that a protester is driven to participate if she expects more people similar to herself to join the protest but would rather stay out if she expects predominantly distant groups to participate.

Then, the central focus of this paper is on understanding how the expected turnout within an individual's own group, as well as in other groups, influences their willingness to protest in contexts where protests involve heterogeneous groups. Moreover, we also examine how these influences are shaped by the agents' existing biases about participation by each group. In the remainder of this section, we discuss potential hypotheses and channels that might affect these phenomena.

2.1 Misperceptions about Other Groups Behavior

It is impossible to characterize the role that the expected turnout of different groups has on the decision to protest without studying agents' expectations about other groups' turnout. There is vast evidence showing that people misperceive others' behavior. Bursztyn and Yang (2022) present a review of this literature and show that these misperceptions are more pronounced when they are about out-group members.

One of the main causes of this misperception is the presence of stereotypes about the behavior of given groups. There are two main approaches to the formation of stereotypes. First, the social psychology approach sees stereotypes as oversimplifications of group traits that may reflect the internal motivations of the person who defines these stereotypes (Macrae and Bodenhausen, 2000; Schneider, 2005). These stereotypes are generally rooted in history and past experiences. In our framework, this could imply that higher-income groups, historically tied to the upper-class and right-wing parties, are attached to lower participation in pro-social activities than their lower-income counterparts. Then, their participation is expected to be underestimated.

Second, the cognition approach proposes that stereotypes arise from cognitive theories that people use to save cognitive resources in their daily lives (Schneider et al., 1979; Hilton and Von Hippel, 1996). Bordalo et al. (2016) link these ideas to the theory of Kahneman and Tversky (1972): stereotypes arise because people use heuristics when computing probabilities, and they use observed behavior as cues to estimate these probabilities. In that sense, the less information one has about the group of people under concern, the more likely to present biases in the estimated probabilities. Some factors might worsen the biases in stereotypes. Segregation, for instance, decreases the information agents have about others' behavior and increases the salience of certain social categories (Enos, 2017), increasing the differences in biases for the out-group and the in-group (Allport, 1954). In our framework, this approach would explain stronger in-group bias in communities or schools that have a higher level of segregation or whose members belong predominantly to one group.

2.2 Strategic Incentives to Participate in Protests

In general, the literature has shown evidence of both strategic complementarity and substitutability. Complementarity might result from the idea of power in numbers (DeNardo, 2014),

social image concerns (Enikolopov et al., 2020), or the social-event nature of protests (Uhlaner, 1989). Substitutability might arise from incentives to free-ride, which is expected to be the case when, for instance, the goal of the event is modest (Tullock, 1971; Shadmehr, 2018). In what follows, we discuss the two types of heterogeneities in strategic incentives that can arise. First, conditional on the group agents belong to, they might present different strategic incentives with respect to in-group and out-group members. And second, citizens belonging to different groups in society might behave inherently differently.

2.2.1 In-group vs Out-group Strategic Incentives

There are several reasons why one would expect a positive differential between the effect of in-group and out-group expected turnout. Here, we delve into two factors that may be influential: (i) social connection to the movement varies with the groups that participate, and (ii) the value of a successful movement depends on the people who organize and participate.

Social connection to the movement is critical for people's decision to join a movement. First, when individuals know protesters, they might feel more connected to a movement and obtain a higher social value of participating (Snow et al., 1980; Uhlaner, 1989). Second, individuals might feel a stronger need to reciprocate protesters when they feel more connected to the groups participating (Lubell and Scholz, 2001; Fehr and Gächter, 2000). Finally, agents may present stronger social image concerns with respect to agents belonging to their own group.

The outcome of the movement also plays a role in the agents' decision to participate. When the movements' demands and the subsequent bargaining process are endogenous to those who participate, the value that a potential activist expects to obtain from the movement might depend on the groups who participate. Then, if agents' preferences correlate to the group they belong to, then the value of joining the movement increases with the expected turnout of the in-group.

2.2.2 Strategic Incentives across Groups

In the previous section, we analyze the distinction between in-group and out-group strategic behavior. An additional point is whether this behavior is uniform across groups. Are there some specific features of groups that make them more likely to respond to incentives in one way or another? It might be that some groups may have stronger social motives or other-regarding preferences. In this line, Hager et al. (2022) study an experiment in protests and counter-protest

interactions and find heterogeneity in responses by left and right-leaning groups. As turnout increases, participation by the left increases—i.e., strategic complements—and participation by the right decreases—i.e., strategic substitutes. A possible explanation for this would be that the left has stronger social connections to the movement.

3 Experimental Design

We design a field experiment to affect agents' beliefs about different groups' participation and study the effect of these beliefs on the decision to protest. We implement this experiment on undergraduate students from two universities in Chile, in the context of the Womens' day March of 2023.

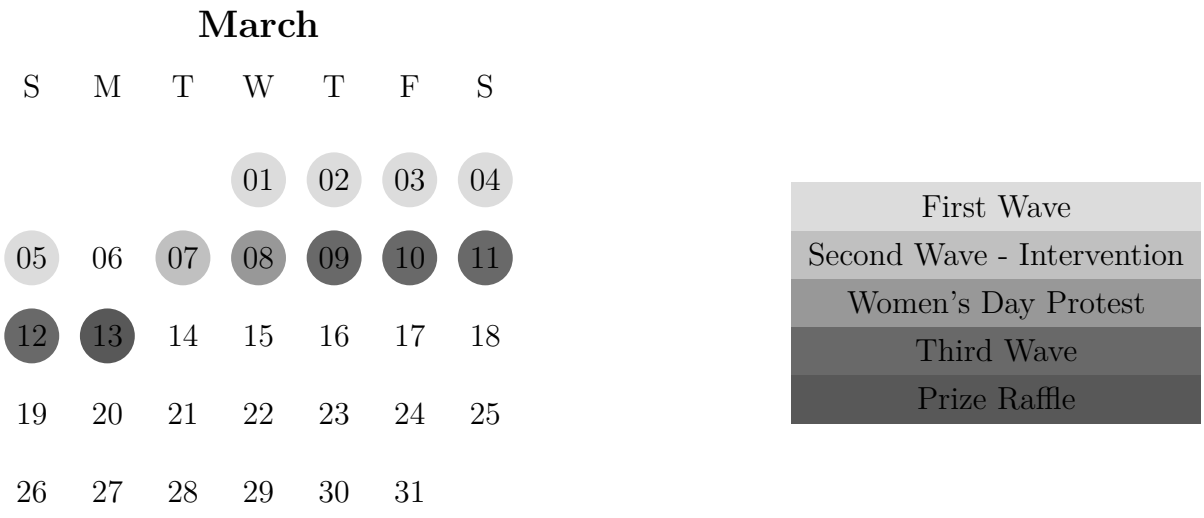
Women's Day demonstrations take place every year on March 8th, to commemorate International Women's Day. There are several characteristics that make this march a good case study. First, it is one of the biggest protests held yearly in Santiago (historic participation rates can be found in Table D-1). Second, it is particularly known for being peaceful, so the risks to participants of the experiment are very low. And third, it unites people of very different backgrounds: ideologies, social and economic status, ages, etc.[3] This specific heterogeneity in participation allows us to explore how different groups of potential protesters interact strategically.

3.1 Survey Design

The experiment is implemented through an online survey that comprises three stages: a Baseline Survey, where we collect students' basic information and intentions to participate in the demonstration; an Intervention, where we deliver information about the expected participation from different groups; and a Follow-up Survey, where we obtain reports on actual participation, together with additional demographic information. The intervention is the critical stage of the experiment, in which subjects are randomly assigned to different treatments that consist of information about different groups. In this way, we are able to track agents' beliefs, participation, and the effects of different information treatments on these. Figure 1 shows the timeline of the intervention.

[3]See Table D-4 for a characterization of the experimental sample and its perceptions of the nature of this protest.

Figure 1: Experiment Calendar: March 2023



3.1.1 Baseline Survey (March 1 - 5)

The Baseline Survey was conducted the week before the march (March 1-5). It gathers general information about participants, such as sociodemographic characteristics, past participation in the movement, opinions with respect to feminism and sexism, ideological position, religion, and beliefs about expected participation at the upcoming Women's Day demonstrations (COES: Reproducible Research and Studies, 2023; Paredes et al., 2023).

3.1.2 Intervention (March 7)

In the Intervention, we use the data we gathered in the first wave and deliver information about the estimated participation of different groups according to the treatments we specify in detail below. Once the treatment is received, we ask students to elicit again their intention to participate and subsequent beliefs about turnout at the march.

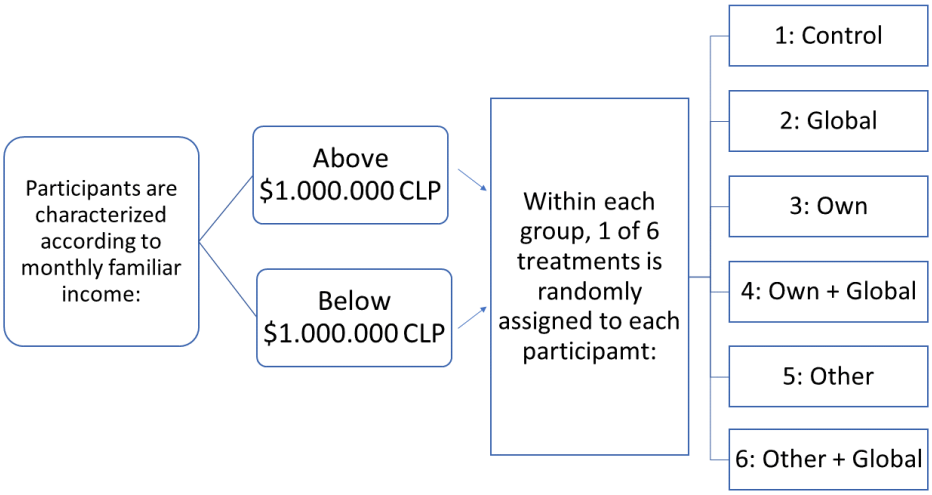
In this stage agents are randomly assigned to treatments in which we deliver relevant information. Two types of relevant information were (potentially) provided: (i) information on overall expected participation at the march and (ii) information on turnout of a specific group of potential participants. In this way, each person received a combination of information about global participation and/or participation of some group to which they might belong or not. Hence, there are six possible treatments:

Treatments:

1. Control: No information is provided.
2. Global: Information on global participation is provided.
3. In-group: Information on own group participation is provided.
4. In-group + Global: Information on own group and global participation is provided.
5. Out-group: Information on the other group's participation is provided.
6. Out-group + Global: Information on another group's and global participation is provided.

Information nudges are constructed using the intention to attend that agents reported in the Baseline Survey (Table 1). We refer to these as "true" priors in the rest of this paper, although they are by no means the effective participation rates. These are simply the only internally consistent statistics available to use as possible "true" parameters for the experiment.

Figure 2: Survey Structure



Naturally, students are heterogeneous along several dimensions: ideology, religion, cohorts, income levels, etc. We decided to define groups according to their self-reported socioeconomic status, as can be seen in Figure 2. The main reason for choosing this dimension is that the distribution of income across the sample allowed us to divide students into two groups—high and low income—that can be easily interpreted and commonly understood.[4]

[4]The analysis of this paper was preregistered on the XXX repository. Because of the nature of the experiment,

Table 1: **Information used as treatments**

Type of Treatment	University 1	University 2
Global	68%	70%
Relative: Below $ 1MM	67%	73%
Relative: Above $ 1 MM	70%	65%

To further illustrate how treatments worked, in Figure 3 we present an example of the information delivered in the Intervention, where the *Treatment 4 - In-group + Global* for lower-income students of University 1 is displayed.

Figure 3: Information delivered for Treatment 4 - University 1 & Lower Income Group Students

> Next we will ask you to predict attendance at the Women's Day march. Some of these questions were already included in the first wave, so before each question we will remind you of your previous answers.
>
> You may have since changed your mind, so we want to give you the opportunity to respond as best as possible. If you continue believing the same as before, there is no problem, you should simply indicate the same number.
>
> We inform you that in the first wave of the survey, **68%** of the people who responded from University 1 reported that they had intentions to participate in the march.
>
> Additionally, **67%** of people from University 1 whose household income classifies as *Under $1,000,000 CLP* reported an intention to participate in the march. You had previously predicted x%.

Randomization of treatments was done automatically by the survey server (Qualtrics) once participants entered the survey. The randomization was programmed to assign treatments in a roughly equal manner across all respondents within the randomization unit: group (income level) and university. Figure C-1 shows the distribution of treatments.

3.1.3 Why Studying Heterogeneity across Income Groups?

Determining the dimension that best captures student heterogeneity is not obvious. A good measure of heterogeneity should discern differences among students based on identifiable characteristics. In that sense, ideology and income are natural candidates. When measured correctly,

the pre-analysis plan did not specify the dimension of heterogeneity that would be used to define reference groups. Instead, it specified the experimental design for a given dimension—i.e., ideology, income, or any other suitable variable—which would then be determined in such a way that allowed balanced treatments in the intervention.

both dimensions can generate groups with recognizable traits and behaviors, which society might attribute either because of prejudice or historical reasons. For example, high-income students might suffer the stereotype of being more right-wing and less politically involved than their low-income peers. Left-wing students might be seen as more active and, hence, more likely to participate in demonstrations than right-wing students. Both of these differences in behavior are good enough to capture heterogeneous strategic incentives in the decision to participate in the Women's March. Hence, the critical factor in deciding a measurement group will be the capacity to measure each dimension correctly. As we argue below, the skewness in the distribution of ideology across students makes it impossible to generate two separate recognizable groups. In contrast, the heterogeneity in students income levels does allow us to separate the high and low-income students in a more consistent way.

To use political ideology as the dimension of heterogeneity, one needs to be able to separate students into (roughly) equally-sized groups that are recognizably different: the left and the right. However, there are two problems with this. First, the measure of ideology is self-reported, and therefore, students located on the left of the distribution might not necessarily identify themselves with the left. Second, the distribution of ideology across students is very skewed to the left, with a median of 3 on an ideology scale from 0 (left) to 10 (right). When receiving information treatments, this could affect results due to misaligned in/out-group identification. In order to avoid such issues, income level was chosen as it is not subject to such interpretation problems.

To gather information about income level, subjects were asked about identification with a social class (low, medium, or high), municipality of residence, and monthly household income. As the latter measure is less subject to the interpretation and distribution issues we previously discussed, we chose it as our measure of income. In addition, monthly household income correlates positively with the other socioeconomic variables mentioned (see Table 3), correctly capturing general socioeconomic characteristics.

Students were asked to report monthly household income levels within ranges of $500.000 CLP from $0 to +$2.500.000 CLP (the distribution of students within each university according to income level can be observed in Figure E-2). This income variable was used to divide students into roughly equally sized groups: below $1.000.000 CLP and above $1.000.000 CLP (Figure E-3). This cutoff was chosen instrumentally for the experiment (for reference, the average monthly household income in Chile for the year 2022 was $1.304.771 CLP (Ministerio

de Desarrollo Social y Familia, 2022)). To avoid confusion, when we deliver information treatments and refer to groups throughout the experiment, we explicitly refer to groups as "Below $1.000.000 CLP" and "Above $1.000.000 CLP".

3.1.4 Third wave (March 9 - 12)

The third wave of the survey was implemented after the protest. In this stage, we obtain subjects' reports on their actual participation in the march and their motives behind that decision.

3.2 Experimental Sample

Our experimental population consists of undergraduate female students from two Chilean universities (from now on referred to as University 1 and 2). This sample was chosen for two reasons: on the one hand students have had an important historical role in Women's Day protests, and on the other hand it allows us to reach a relatively large number of individuals, while keeping control of the experiment and sample.

All surveys were conducted online using the Qualtrics platform. Invitations were originally sent to all undergraduate students from institutional university e-mails. During stages two and three, invitations were sent from an e-mail account generated specifically for this experiment. Reminders were sent via e-mail and text message (for those who provided their phone numbers). In order to encourage participation, tickets to a popular music festival were raffled among subjects who participated in all three waves of the survey.

Thanks to the agreed participation of the two aforementioned universities, around 2.944 students participated in the first wave, of which 2.053 were considered due to initial filters (for details, see Appendix Table D-2). Of these 2.053 students, 1.765 finished the first survey (see Table 2).

Although there was a high response rate during the Baseline Survey, the second wave suffered from attrition. To evaluate if this attrition challenges the internal validity of the experiment, we run regressions to determine if attrition was not random. Table D-3 shows that most characteristics are not statistically significant in explaining participation in the second stage, with the only exception of the year of admission to university. We thus claim that selective attrition is not a threat to internal validity.

If we take final participation and compare to the total number of female students in each

Table 2: **Participation by Survey Wave**

	Baseline Survey	Intervention	Follow-up Survey
University 1	1403	1087	1036
University 2	362	312	302
Total	1765	1399	1338

university, we find that University 1 had a final response rate of 4.4% and University 2 of 2.8%.[5] Power analyses indicate that the resulting sample sizes are sufficient in order to achieve at least 80% power (see Figure B-1).

3.3 Sample Characterization

We characterize the final experimental sample for determining any underlying imbalances in observable characteristics (for a general review of outcome variables, see Table D-4). Table 3 studies if there are any differences in observable characteristics between the income groups considered. There are statistically significant differences between income groups in the ideological position, justification of violence, protest violence and repression expectations, household income range, social class, age and mother's work status. None of these are surprising, as they go in line with expected differences between higher and lower income groups due to social and economic stratification. The higher income group, on average, has a more right-inclined political ideology, higher social class identification and higher chances of having a working mother when young.

Differences that are significant and interesting to highlight are that the higher income group displays, on average, a lower justification of violence and lower expectations of repression and violence on Women's Day Protests. Meanwhile, the studied income groups had no significant differences in their levels of interest in politics, experienced discrimination, average past participation in marches, and appreciation of the feminist movement. These results are similar to Castle et al. (2018), who find that people who were most likely to be aware and mobilized by the #MeToo movement were Democrats, those with higher political interest, and those who have experienced sexual harassment, but find no significant relationship between awareness and mobilization with income level.

When analyzing the sample's balance across treatment groups, we found few significant dif-

[5]University 1 has 45.248 undergraduate students, where 23.403 students (51.72%) declare to be women. On the other hand, University 2 has 20.762 undergraduate students, of which 10.868 are women (52.35%) (Mifuturo, 2023).

Table 3: **Characterization of Income Groups**

	Lower Income Group		Higher Income Group		Difference
	Mean (1)	Sample (2)	Mean (3)	Sample (4)	in means (5)
Ideological Position(1=Left, 10=Right)	2.94	794	3.22	543	-0.29***
Level of interest in Politics	2.34	788	2.38	541	-0.04
Frequency of conversation about Feminist topics	3.45	795	3.45	542	-0.00
Appreciation of Feminist Movement	4.49	795	4.42	543	0.07
Movilization leads to more change	4.43	787	4.31	537	0.12***
Violence is justified in some cases	2.82	773	2.53	528	0.28***
Women's movilization has led to positive changes	4.40	788	4.35	540	0.05
Expected level of violence (1=Peaceful, 5=Violent)	1.83	794	1.75	543	0.08*
Probability of repression in Women's Day Protests	52.39	795	47.07	542	5.32***
Income group identification	2.04	776	3.87	529	-1.83***
Monthly Household Income Range	1.48	795	4.36	543	-2.88***
Age	20.93	794	21.35	542	-0.42**
Have felt discriminated	0.85	717	0.82	487	0.03
Mother worked for salary when I was young	0.79	747	0.87	529	-0.07***
Average participation in marches 2019-2022	0.39	787	0.41	542	-0.02
Non Traditional Gender Role Attitude	4.06	795	4.18	543	-0.13***
Proactive Gender Role Attitude	4.28	795	4.23	543	0.05
Believes women are discriminated in life	4.55	795	4.52	543	0.03

Notes: Table shows the mean and sample size of several studied characteristics by income group. Column (5) reports the difference between means, and the significance of a T-Test of means. The analyzed sample considers both universities. * p<0.10; ** p<0.05; *** p<0.01.

ferences (see Tables D-5, D-6 and D-7). Among these, the only variable that seems to have a relevant imbalance, as it appears for both income groups and also has a significant F-test, is having a working mother when young. This could affect results if having a working mother directly relates to protest participation, which could be a possibility if class struggles are associated with protest demands, for example. Still, as we control for income group by construction, in addition to other SES variables and political ideology, results should not be biased by this particular imbalance.

In addition, University 2 had more sample imbalance due to lower participation and a smaller number of observations (Table D-7). To account for this, our estimations include controls for the variables that show to have imbalances (interest in politics, ideological position, appreciation of the feminist movement, and age). Altogether, the results of these multiple balance tests indicate that treatment randomization was successful in generating balanced treatment groups.

4 Results

In this section, we present the main results of the paper, divided into three sets of results. First, we show evidence of strong and heterogeneous misperceptions about others' intentions to participate. Second, we provide evidence of the effect of the implemented treatments over agents' posteriors beliefs, and third, we show evidence concerning strategic incentives to participate. We finish this section with a discussion of the mechanisms behind our main results.

4.1 (Mis)perception about Others' Intention to Participate

Prior to any analysis of the role of beliefs in the decision to participate, it is crucial to understand these beliefs. To do this, we study how far away agents' predictions about others' intentions are from their true values.

Table 4 illustrates agents' beliefs about the intentions to participate of different groups—i.e., the global pool of potential participants, the lower-income, and the higher-income—and compares them to the true values, as reported by the same students. To construct this (and to construct our treatments) subjects were not only asked about their own intention to participate, but also about their beliefs about others' intentions. The exact question was: *From each of the following socioeconomic groups, what do you think is the percentage of female students at your university who intend to participate in the march?*

Panel A of Table 4 shows mean beliefs for students in university 1, while Panel B does the same for university 2. Each row corresponds to a specific reference group, and for each of them, we report the true value of the percentage of subjects who intend to participate in the march, the lower-income group's beliefs about this value, the higher-income group's beliefs, the difference between both groups' beliefs, and the respective p-value.

Notice, first, that global participation was overestimated by about 2 percentage points in University 1, and it was underestimated by about 4 percentage points (see Figure C-2 for a distribution of priors). In both universities, these distortions in beliefs seem universal, as there are no significant differences between income groups.

Secondly, the intention to participate in both income groups is, on average, underestimated. However, two results are worth highlighting. The underestimation of the intention to participate of higher-income students is remarkable compared to that of lower-income students. In University 1, lower-income students' beliefs about the intended participation of high-income

Table 4: **Beliefs about Others' Intention to Participate**

		Mean Belief			
	Truth	Lower Income Group	Higher Income Group	Difference (2)-(3)	P-value
	(1)	(2)	(3)	(4)	(5)
Panel A: University 1					
Global attendance prior	68	69.64	70.20	-0.56	0.53
Lower income attendance prior	67	65.46	65.67	-0.20	0.84
Higher income attendance prior	70	47.33	53.41	-6.08***	0.00
Panel B: University 2					
Global attendance prior	70	66.34	66.00	0.34	0.87
Lower income attendance prior	73	64.38	59.24	5.14**	0.02
Higher income attendance prior	65	45.23	51.48	-6.25**	0.01

Notes: Column (1) displays subjects' reported intention to participate. Columns (2) and (3) display average beliefs by income group before receiving treatment. Column (4) shows the difference in belief of both income groups, and column (5) the p-value of the t-test of means. * p<0.10; ** p<0.05; *** p<0.01.

students are about 23 percentage points below the true value. But notice that it is not only the lower-income students who misperceive: the high-income students underestimate their own group's intentions to participate by about 17 percentage points.

Finally, the intention to participate of the lower-income group was also underestimated but notably less than higher-income attendance. In University 1 both income groups predicted attendance of this group at around 65% while the truth was 67%, only 2 pp in difference. In University 2 there is a greater underestimation: the lower income group predicted on average an attendance of 64% while the higher income group an average of 59% and the truth was 73%, 9-14 pp in difference respectively. Here it is important to note that University 1 comprises 77% of the sample, so when considering treatment effects one must take into consideration that on average the truth of lower income attendance was not that far from average attendance beliefs.

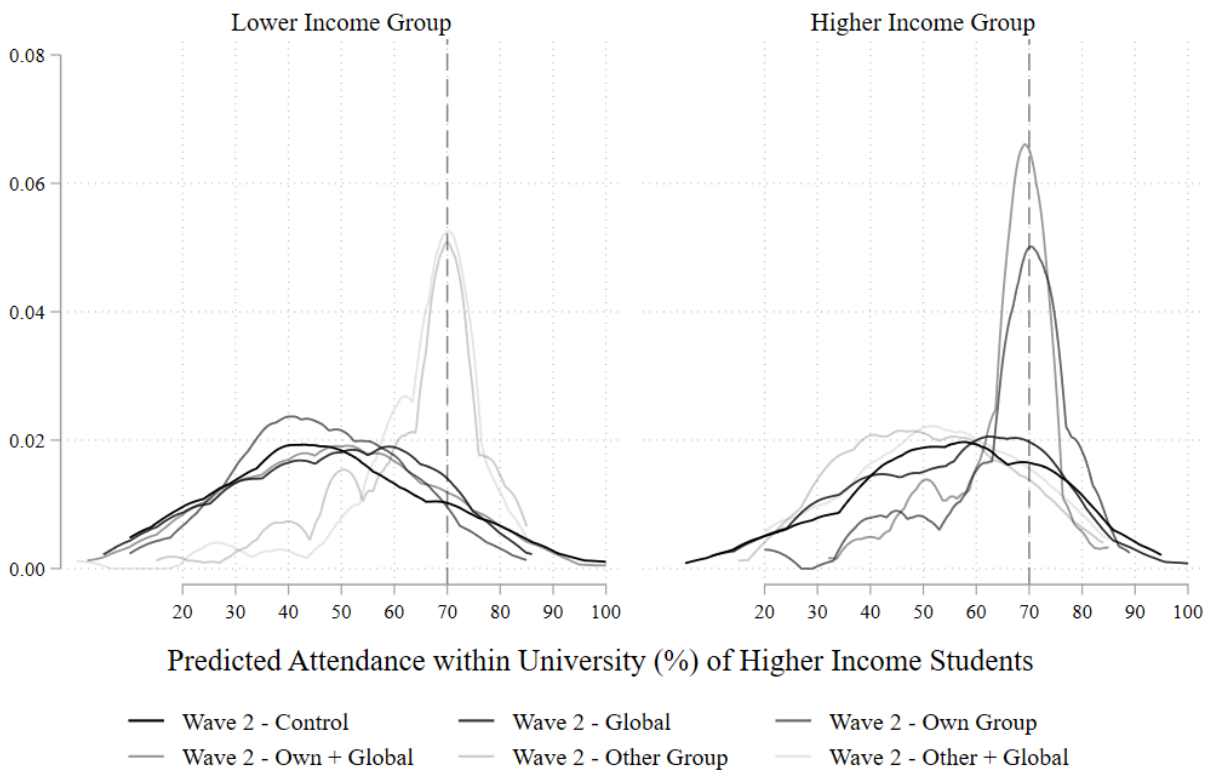
4.2 Treatment Effects on Posterior Beliefs about Participation

In this section, we study how treatments affect posterior beliefs on people's turnout. Two properties are fundamental for treatments to be considered successful: (1) treatments are capable of moving posterior beliefs, and (2) the direction in which people update their beliefs must be rational, i.e. if their priors were above the information treatment, they tend to update negatively

towards the truth, and inversely if they were below the information treatment.

Figure 4 shows density graphs for posterior beliefs about participation of the higher income group of University 1 (see Figures C-3 and C-4 for other posteriors of University 1, and C-5 - C-7 for University 2). Treatment groups including information about the true planned participation of the higher income group (*Other* and *Other+Global* for the lower income group, and *Own* and *Own+Global* for the higher income group) present a distribution that is more compressed around the treatment than the control group and other treatments.

Figure 4: Posteriors about Participation of Higher Income Group (University 1)



Notes: Kernel density estimates are displayed for higher income attendance beliefs for both income groups (shown in different panels) and all treatments (different colored lines). Sample considers only University 1 and beliefs are posteriors, i.e. post receiving the information treatment. Vertical dashed line is the treatment truth delivered.

To complement this, we run the following first-stage regressions:

$$\text{Posterior Belief}_i = \alpha_0 + \tau_1 \text{Global} + \tau_2 \text{In-Group} + \tau_3 \text{Out-Group} + \theta X_i + \epsilon_i \tag{1}$$

where Posterior Belief$_i$ is the belief of individual i on either global, in-group, or out-group participation. Global, In-Group and Out-Group are indicator variables for the treatment received,

and X_i a vector of variables that include university, value of the feminist movement, characterization of the protest (more peaceful or violent), political ideology, interest in politics, past participation in Women's Day marches, commune of residence and participation priors (global and relative). As treatment's effects on posterior beliefs will depend on agents' prior beliefs, each regression is run in two sub-samples, which divide agents among those whose prior beliefs lie above the true reported intention to participate and those whose prior beliefs lie below this true value. The intuition for this separation is that these two samples should update in opposite directions.

Table 5 shows three sets of estimations depending on the posterior beliefs analyzed: global participation (columns 1 and 2), participation of the lower-income group (columns 3 and 4), and participation of the higher-income group (columns 5 and 6). Odd columns show results for the subset of agents whose priors were below the true reported intention to participate, and even columns show those who were above. We report only the coefficient corresponding to the posterior analyzed, but in all regressions, we control for the other treatments.[6]

The results show that overall treatments had an effect on posterior beliefs in all groups, with the exception of the Out-Group treatment when students had a prior above the truth (column 6). The Global treatment had an effect of about 2 percentage points depending on whether subjects were above or below truth (columns 1 and 2). This can be attributed to the small differences between priors and treatments observed in this case. For the lower-income treatment, the effect was about 9 percentage points for those below truth (column 3) and about 6 percentage points for those above truth (column 4).

As discussed above, a majority of students underestimated the intention to participate of the higher-income group. This explains why the treatment of this group is the most powerful: it increases the posterior belief for this group by about 20 percentage points for lower-income students and by 13.5 percentage points for higher-income students who were below the truth (column 5). As most subjects underestimated higher-income group participation, the subsample of subjects with priors above the true value has limited power (133 observations versus 300-600 observations in the other subsample/columns) and, therefore, displays no significant effects (column 6). These estimated treatment effects are robust to controlling for subject characteristics and prior beliefs.

[6]Since we have treatment arms with 2 treatments, we have run alternative specifications where we include treatment interactions. The results are shown in appendix table D-8 and are quantitatively similar to those in table 5.

Table 5: **First Stage: Treatment Effects on Posterior Beliefs**

	Participation Posteriors					
	Global		Lower Income		Higher Income	
	Prior Below Truth (1)	Prior Above Truth (2)	Prior Below Truth (3)	Prior Above Truth (4)	Prior Below Truth (5)	Prior Above Truth (6)
Panel A: Lower Income Students						
Global Treat.	2.744**	-1.845**				
	(1.330)	(0.757)				
Own Treat.			8.888***	-5.899***		
			(1.449)	(1.274)		
Other Treat.					19.798***	-2.667
					(1.404)	(3.130)
Observations	317	467	419	365	651	133
Panel B: Higher Income Students						
Global Treat.	2.409*	-2.688***				
	(1.354)	(0.860)				
Other Treat.			8.839***	-6.602***		
			(1.959)	(1.359)		
Own Treat.					13.547***	-0.761
					(1.582)	(2.959)
Observations	215	326	276	265	412	129

Notes: In each regression all treatment variables are included. Regressions also include the following controls: university dummy, value of feminist movement, protest is characterized as peaceful/violent, political ideology, interest in politics, past participation in Women's Day marches, commune of residence and attendance priors (global and relative). Robust standard errors are reported in parenthesis. Both universities are included in this analyzed sample. * p<0.10; ** p<0.05; *** p<0.01.

4.3 Strategic Decision to Participate

We now proceed to analyze whether changes in posterior beliefs had an effect on intended and effective participation in the protest. First, we analyze the direct link between posterior beliefs and participation, and then we use the treatments as instrumental variables for posterior beliefs.

4.3.1 Second Stage: Posterior Effects on Participation

We estimate the following model:

$$\text{Participation}_i = \alpha_1 + \beta_1 \text{Posterior Belief}_i + \theta X_i + e_i \tag{2}$$

where Participation_i can be either the intention to participate—measured as the probability to participate in the protest—or effective participation reported in Wave 3 of the Survey. $\text{Posterior Belief}_i$ is any of the three posterior beliefs (global, higher or lower income), and X_i is the same set of control variables described in the previous section. The results are shown in Table 6. Columns 1-4 show results for the sample of lower income students, while columns 5-8 do the same for higher income students. Odd columns show results for students whose prior belief was below the average participation plan, while even columns do the same for students with a prior above the average participation plan.[7]

We find that global participation beliefs have a positive and statistically significant effect for lower-income students who had priors below the truth. The magnitude of these effects appears to be small: an increase of 1 percentage point in global participation beliefs increases the reported probability of lower-income students to participate in the march by 0.2 percentage points, and the effective participation by 0.5 percentage points. Regarding in-group beliefs, we only find statistically significant effects on higher income students with priors below the truth. On the contrary, we find an effect of out-group beliefs for lower income students. These results indicate complementarities in the strategic decision to participate in protests, both with the global population and the out-group, in the case of lower-income students, and with the in-group for higher-income students.

4.3.2 Two Stage Estimates (IV)

Changes in posterior beliefs could come from various sources—such as from discussions in online forums about the feminist movement in the week prior to the march—which can also have an effect on the actual participation in the march. To account for this, we implement two-stage estimates using the treatments as instrumental variables for the change in posteriors. In the first stage, which was discussed in Section 4.2, we estimate the effects of treatments on subject's posterior beliefs of participation. In the second stage, we exploit variation in beliefs induced

[7] In Table D-9 in the appendix we show results without splitting the sample into these groups.

Table 6: **Posterior Effects on Participation**

	Sample: Lower Income Students				Sample: Higher Income Students			
	Participation Probability		Effective Participation		Participation Probability		Effective Participation	
	Prior Below Truth (1)	Prior Above Truth (2)	Prior Below Truth (3)	Prior Above Truth (4)	Prior Below Truth (5)	Prior Above Truth (6)	Prior Below Truth (7)	Prior Above Truth (8)
Panel A: Global Participation Beliefs								
Global Posterior	0.297**	-0.060	0.005**	-0.003	-0.413	0.206	-0.004	0.002
	(0.140)	(0.178)	(0.002)	(0.003)	(0.276)	(0.244)	(0.004)	(0.004)
Mean of Dep. Var.	52.55	60.39	0.36	0.46	57.15	62.05	0.45	0.48
St. Dev. of Dep. Var.	34.63	34.37	0.48	0.50	36.25	34.78	0.50	0.50
Observations	316	467	311	464	214	326	213	324
Panel B: In-Group Participation Beliefs								
Lower Income Posterior	-0.042	-0.108	-0.002	-0.001				
	(0.116)	(0.152)	(0.002)	(0.002)				
Higher Income Posterior					0.188*	-0.141	0.001	-0.002
					(0.101)	(0.258)	(0.002)	(0.005)
Mean of Dep. Var.	54.02	60.88	0.39	0.46	58.10	66.50	0.45	0.54
St. Dev. of Dep. Var.	34.61	34.41	0.49	0.50	35.71	33.83	0.50	0.50
Observations	418	365	413	362	411	129	410	127
Panel C: Out-Group Participation Beliefs								
Higher Income Posterior	0.140**	0.126	0.001	0.001				
	(0.066)	(0.186)	(0.001)	(0.003)				
Lower Income Posterior					0.013	-0.167	0.001	-0.004
					(0.132)	(0.212)	(0.002)	(0.003)
Mean of Dep. Var.	55.57	65.32	0.40	0.54	56.70	63.64	0.42	0.52
St. Dev. of Dep. Var.	34.67	33.60	0.49	0.50	35.86	34.68	0.49	0.50
Observations	650	133	642	133	275	265	273	264

Notes: Regressions include the following controls: university dummy, value of feminist movement, protest is characterized as peaceful/violent, political ideology, interest in politics, past participation in Women's Day marches, commune of residence and attendance priors(global and relative). Robust standard errors are reported in parenthesis. Both universities are included in this sample. * p<0.10; ** p<0.05; *** p<0.01.

exogenously by treatments to estimate the final effect of beliefs about participation on the individual's own decision to participate in the demonstration.

Each treatment is included in a separate regression. Therefore, for a given group $Z \in$

$\{Global, In-Group, Out-Group\}$, we estimate the following IV regression:

$$\begin{aligned} &(1) \quad \mathbf{Z\ Belief}_i = \alpha_0 + \tau_1 \text{Treatment} + \theta X_i + \epsilon_i \\ &(2) \quad \text{Participation}_i = \alpha_1 + \pi_1 \widehat{\mathbf{Z\ Belief}}_i + \phi X_i + e_i \end{aligned} \tag{3}$$

As before, we estimate equation 3 in the subsamples of students whose priors were either below or above the truth. The results are shown in Table 7. Panel A shows the results for changes in global participation beliefs, while Panel B (C) shows results for in-group (out-group) beliefs. Columns 1-4 considers lower income students, while columns 5-8 considers higher income students. The first stage is reported in Table D-10 of the Appendix.

The IV estimates confirm our previous findings of lower income group complementarities with higher income group participation in the demonstration. In addition, estimations also show evidence of complementarities for the higher income group with both the in-group and the out-group's participation.

In Panel A we see that the global treatment is a weak instrument despite relatively balanced subsamples (200+ observations per column), therefore we refrain from interpreting these estimates causally.

In Panel B the instrument performs well in all cases except columns 6 and 8 (higher income students whose priors were above the truth). We find effects for the in-group posterior only on higher income students: an increase of 1 percentage point in the belief of higher income group participation increases effective participation by 0.6 percentage points and the probability to attend by 0.43 pp (at a 5% significance level).

Finally, the instrument performs well in most regressions of Panel C, with the exception of columns 2 and 4 where we have few observations. We find evidence of complementarity between beliefs about higher income students' participation and participation for both income groups. An increase in 1 pp in beliefs about higher income group participation increases the probability of participation by 0.2 pp and effective participation by 0.4pp for the lower income group. Regarding higher income students, an increase of 1 pp in beliefs about lower income group participation increases their probability of participation by 1 pp and effective participation by 1.8 pp. The effect is large, since a change of 1 s.d. in the posterior increases effective participation in 0.36 s.d. The effect is also large when compared to the findings in similar studies. For instance, Cantoni et al. (2019) find that an increase of 1 pp of global attendance beliefs decreases turnout to the protest by about 0.5 pp.

To sum up, our results show strategic complementarity between beliefs about out-group

Table 7: **Two Stage Estimates: Treatment Effects on Participation**

	Sample: Lower Income Students				Sample: Higher Income Students			
	Participation Probability		Effective Participation		Participation Probability		Effective Participation	
	Prior Below Truth (1)	Prior Above Truth (2)	Prior Below Truth (3)	Prior Above Truth (4)	Prior Below Truth (5)	Prior Above Truth (6)	Prior Below Truth (7)	Prior Above Truth (8)
Panel A: Global Participation Beliefs								
Global Posterior	0.177	0.915	0.027	-0.016	-0.494	1.192	0.048	-0.008
	(1.393)	(1.497)	(0.024)	(0.022)	(1.696)	(1.124)	(0.041)	(0.018)
Mean of Dep. Var.	52.55	60.39	0.36	0.46	57.15	62.05	0.45	0.48
St. Dev. of Dep. Var.	34.63	34.37	0.48	0.50	36.25	34.78	0.50	0.50
Observations	316	467	311	464	214	326	213	324
Kleibergen-Paap F Statistic	2.47	5.58	2.64	6.38	2.58	9.01	2.37	9.14
Panel B: In-Group Participation Beliefs								
Lower Income Posterior	-0.029	1.024	0.000	0.005				
	(0.419)	(0.692)	(0.006)	(0.010)				
Higher Income Posterior					0.431**	-0.688	0.006**	-0.018
					(0.195)	(2.325)	(0.003)	(0.045)
Mean of Dep. Var.	54.02	60.88	0.39	0.46	58.10	66.50	0.45	0.54
St. Dev. of Dep. Var.	34.61	34.41	0.49	0.50	35.71	33.83	0.50	0.50
Observations	418	365	413	362	411	129	410	127
Kleibergen-Paap F Statistic	29.71	19.86	27.63	18.85	100.60	0.58	103.26	0.49
Panel C: Out-Group Participation Beliefs								
Higher Income Posterior	0.216*	1.139	0.004**	-0.032				
	(0.119)	(4.768)	(0.002)	(0.082)				
Lower Income Posterior					0.355	1.071**	0.012	0.018**
					(0.495)	(0.544)	(0.008)	(0.009)
Mean of Dep. Var.	55.57	65.32	0.40	0.54	56.70	63.64	0.42	0.52
St. Dev. of Dep. Var.	34.67	33.60	0.49	0.50	35.86	34.68	0.49	0.50
Observations	650	133	642	133	275	265	273	264
Kleibergen-Paap F Statistic	262.58	0.23	263.03	0.23	17.13	40.86	16.66	41.43

Notes: Regressions include the following controls: university dummy, value of feminist movement, protest is characterized as peaceful/violent, political ideology, interest in politics, past participation in Women's Day marches, commune of residence and attendance priors(global and relative). Robust standard errors are reported in parenthesis. Both universities are included in this sample. * p<0.10; ** p<0.05; *** p<0.01.

participation and participation in the manifestation for both lower and higher-income students. Regarding complementarity between beliefs about in-group participation and own participation, we only find evidence for the higher-income group.

Crucially, our results seem to be driven by changes in posteriors in different samples. Let us focus first on the sample of high-income students, corresponding to columns 5-8. In panel C, we find significant complementarity with respect to the out-group for those whose priors are above the true participation (columns 6 and 8). These are students who were too optimistic about the turnout of low-income students: the experiment moved their beliefs downward, and this discouraged their participation. This is despite the instrument being strong for those below and above the truth.

Conversely, Panel B shows significant complementarity within their in-group for students who had priors below the actual participation rate (columns 5 and 7). Initially, they were too pessimistic about their own group's turnout, and then they increased their participation upon receiving accurate information.

With the low-income group (columns 1-4), we only find complementarity with respect to the out-group (Panel C), and this complementarity is driven by those students whose priors about high-income students' participation were low enough.

4.4 Mechanisms

To explore the mechanisms behind students' misperceptions about the higher-income group, we implemented a follow-up survey in which students were asked to compare both groups on a series of dimensions. The questions range from political ideology (0-10 scale) to questions that aim to capture pro-social values and attitudes, such as stance on abortion (0-5 scale), religiosity, and the frequency with which students visit a neighbor or talk to someone in trouble (0-7 scale).[8]

We find that both groups of students present misperceptions about the behavior of the high-income group on several dimensions. Figure 5 illustrates, for each income group, their responses to a series of questions that aim to capture their political preferences and pro-social behavior. For each dimension, the top darkest bar represents the average response of the respective group in the survey. The second, lightest bar represents the beliefs that students within the same

[8]The follow-up survey was implemented a year after the intervention, and it aimed to understand the differences in students' perceptions of different groups. As for the previous waves, the survey was implemented in Qualtrics. We received answers from 427 students (342 from University 1 and 85 from University 2.

group have about these responses. The third bar reflects the beliefs that students from the other group hold about these responses. Using the terminology from the rest of the paper, the second bar represents in-group perceptions—how lower-income students (top panel) and higher-income students (bottom panel) perceive their own group. The third bar represents out-group perceptions—how higher-income students perceive their lower-income counterparts in the top panel and how lower-income students perceive their higher-income counterparts in the bottom panel.

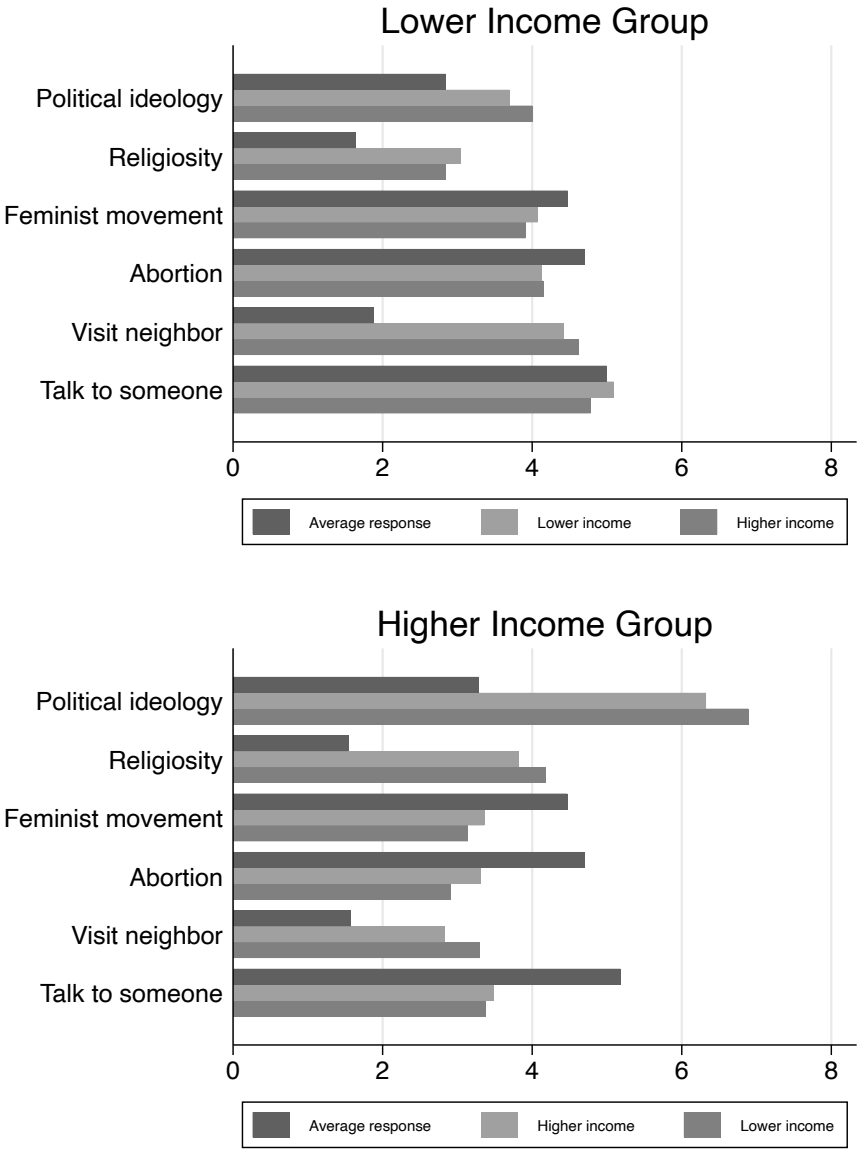
Comparing average responses (dark bar), we find only small differences between both groups, in terms of ideology or in their pro-social behavior. This suggests that there are no intrinsic differences in groups' political and social preferences and behavior. However, both groups are misperceived according to most dimensions, and these misperceptions appear to be stronger for the high-income group. Overall, students' perceptions of religiosity are biased upward, with a stronger bias for the higher-income group. In contrast, preferences for abortion and the value of the feminist movement are underestimated in both groups.

The high-income group is particularly misperceived on two key variables: political ideology and the frequency with which they interact with others (the latter serving as a proxy for pro-social attitudes). Students belonging to this group are perceived as much more right-wing than they actually are, and as less communicative with others. Interestingly, these misperceptions hold true for both low- and high-income students. Contrary to the predictions by Bursztyn and Yang (2022), there seems to be a bias that is more intrinsic to the reference group itself rather than dependent on whether the beliefs pertain to in-group or out-group members.

The question, then, is why are misperceptions stronger for the higher-income group. To see this, we explore the role of students' interactions and their environment in enhancing these misperceptions. Figure 6 shows, for the low-income students, the relationship between the intensity of their interactions with the high-income group and their perceptions about this group for each of the dimensions studied above. In each panel, the horizontal axis shows the percentage of high-income friends students declare to have. The dashed line shows the average response of students from the high-income group, and the continuous line shows the fitted value for students' perception of the high-income group. For all the variables, with the exception of "visiting neighbors", predictions become more accurate as the percentage of high-income friends increases.[9] In general, students who interact less with the high-income group believe they are

[9]Given that there are very few observations at the right end of the horizontal axis, the fit of predictions is noisy at these values.

Figure 5: Groups' Characteristics and Perceptions



Notes: For each measure, the figure shows: (1) the group's average response (i.e., "true value") , (2) the same group's beliefs about the respective group's response, and (3) the other group's beliefs about the respective group's response.

more conservative and present less pro-social behavior than the actual values. In the case of abortion, for instance, most students in the high-income group declare to be in favor of it under all circumstances. However, lower-income students who present unfrequent interactions with the high-income group believe that most of them disagree with abortion. This misperception

decreases as students present more interactions with the higher-income group. The same occurs with religiosity, which tends to be overestimated, and for talking to someone, which is underestimated. In Figure C-8 in the Appendix, we show how higher-income students' interactions with their own group also help correct in-group misperceptions.

The evidence above suggests that students who interact more with higher-income students have a better idea of their behavior, consistent with the relevance of cognitive cues (Hilton and Von Hippel, 1996; Bordalo et al., 2016) and group categorization (Kunda, 1999; Enos, 2017) on the formation of stereotypes. This suggests that students' segregation and the lack of intergroup interactions—either by homophily or by structural reasons—might worsen the precision with which students perceive each other, and this might affect their ability to coordinate into different political activities.[10]

Finally, to understand the mechanisms behind heterogeneity in strategic incentives, we first explore the main drivers of people's incentives to protest. As Figure A-1 in the Appendix shows, we do not find stark differences in participation motives across groups. Priors about global participation seem to be slightly more relevant for the low-income group, as are the beliefs about the relevance of the movement on effective political changes, but these differences would not explain why strategic incentives with respect to specific groups differ. Overall, it seems that changes in the information about specific groups, enhanced by the presence of rooted misperceptions about the high-income group, are the driving forces in the observed heterogeneities.

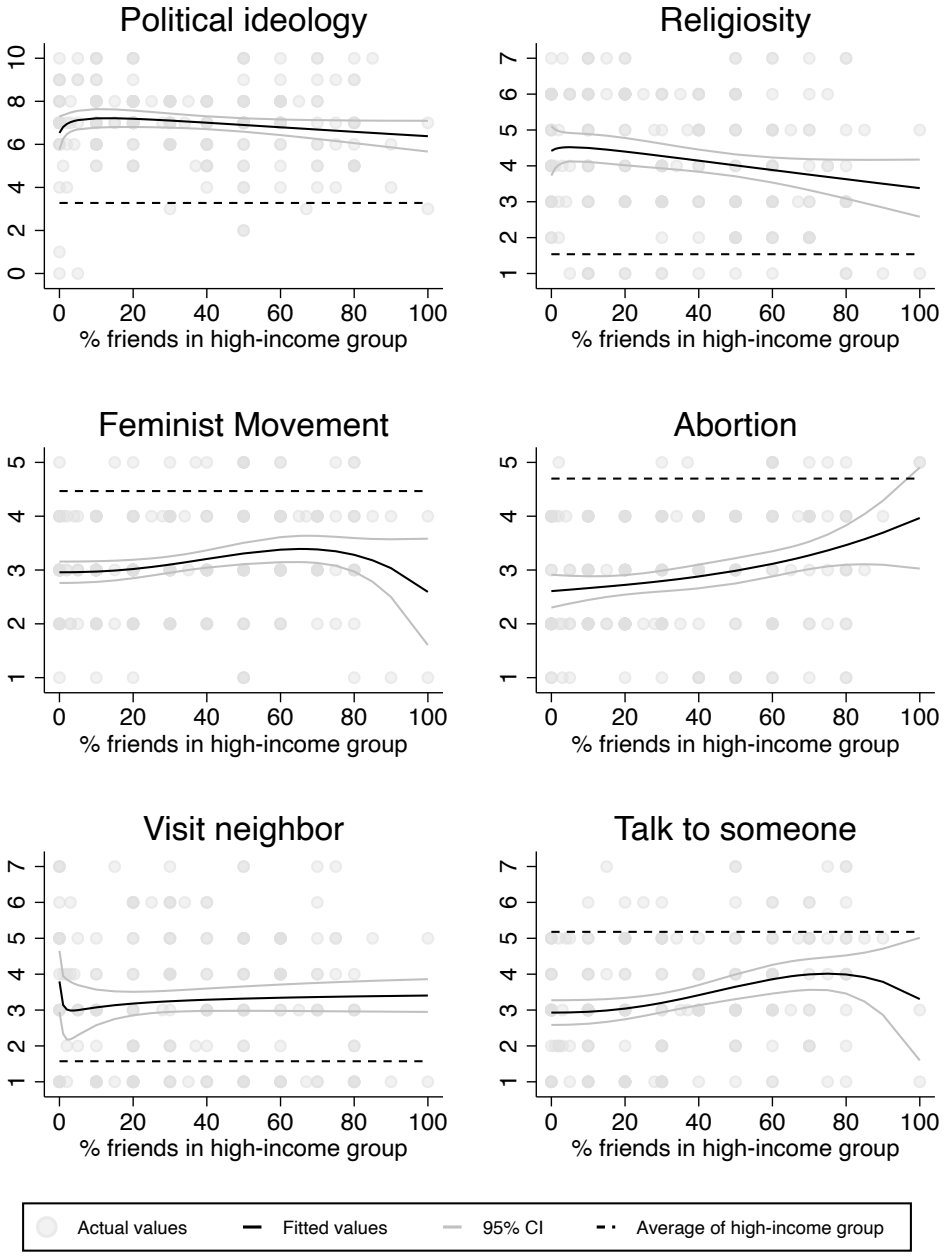
5 Discussion

In this paper, we study heterogeneity in the strategic forces at play within the decision to protest. We implement a field experiment in the context of the Women's March in Chile. Our results highlight the relevance of considering heterogeneity across groups when studying the strategic decision to participate in protests. In particular, we show that individuals not only take into account global turnout in their decision to participate, but also consider the behavior of specific groups of potential participants in different ways.

First, we find evidence of heterogeneity in perceptions about groups' participation. In particular, we show that participation of the high-income group of students is highly underes-

[10]Relatedly, Ananat and Washington (2009) find that increases in racial segregation lead to a decrease in Black civic efficacy, as measured by the ability to elect representatives who support legislation favored by the Black community. See also Corvalan and Vargas (2015) for the effects of segregation on civil conflict.

Figure 6: Students' Interactions and Out-Group Misperceptions: Lower Income Group



Notes: In each panel, the horizontal axis plots the percentage of high-income friends students declare to have, and the vertical axis corresponds to each dimension studied. The dashed line shows the actual average response of the high-income group, and the continuous line shows the fitted value for the lower-income students' perception of the higher-income group's responses.

timated on average by all students, whereas participation of the low-income group seems to be predicted more accurately. In addition, the estimation of in-group participation appears to be

more precise for both income groups.

Regarding strategic incentives to participate, we find that the way agents respond to other people's turnout is heterogeneous across groups. First, even though agents behave as complements, people from different socioeconomic backgrounds behave strategically differently. While the higher income group demonstrates strategic complementarities with both their in- and out-group, the lower income group only behaves as complements with the out-group. Second, we find that a higher turnout of higher-income students has different effects overall than a higher turnout of lower-income students. While a higher turnout by the high-income group increases the expected participation of both the higher and lower-income groups, a higher turnout by the lower-income group appears to be only relevant for the decision of high-income students.

Our results about complementarity—either global or by group— are, in a certain way, puzzling. According to recent literature, Women's Day Marches would be more inclined to encourage substitute strategic incentives due to this protest's more symbolic nature. Shadmehr (2018) postulates that in protests with more material goals, such as toppling regimes, for example, strategic complementary forces dominate, while in more modest/symbolic protests, substitute forces and free-riding concerns would be expected to be greater. In contrast to these predictions, in this experiment we see ample evidence of complementary interactions between attendance beliefs and the decision to participate. Does this challenge the previously mentioned literature? Could it be that in our case, despite the goals being modest, social image concerns and peer effects enter into play, dominating any possibility of free riding? We believe this is up for discussion and in need of further investigation. As mentioned earlier, a possible answer to this puzzle could be that both protests' goals and size should be considered when determining the strategic forces that will dominate.

Also, our results on the heterogeneity of strategic incentives emphasize the importance of considering heterogeneous interactions between different groups of people when studying participation in protests. Traditionally, the literature has focused on the effects of beliefs about the entire pool of potential participants on agents' decision to participate. In this study, we find that relative beliefs—i.e., beliefs about the participation of specific groups—are more critical for participation than global beliefs. Furthermore, we find that the effect of out-group turnout on participation is twice as powerful as the effect of the in-group for high-income students. These results are suggestive of important strategic concerns in the participating decision that have been, in a sense, disregarded: groups who are thought to participate less and/or groups

who have differences or animosities with their own group could have a greater influence on a subject's strategic decision.

We believe that this investigation paves the way for future work regarding heterogeneous interactions in the strategic decision to participate. This paper's findings confirm the existence of complementary forces at work and, most importantly, of different magnitudes depending on the reference group being considered. This is why a further effort must be made to elucidate the mechanisms behind strategic interactions in protest games, particularly taking into consideration and giving room to heterogeneous incentives in different protest contexts.

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Online Appendix

Group (Mis)Perception and Strategic Behavior in Protests: Evidence from Women's March in Chile

(Not for Publication)

Sofía Correa, Agustina Crozier, Francisco J. Pino

A Protest Motives

To further characterize the framework, we explore subjects' participation motives (see Figures A-1 & A-2 and Table A-1).

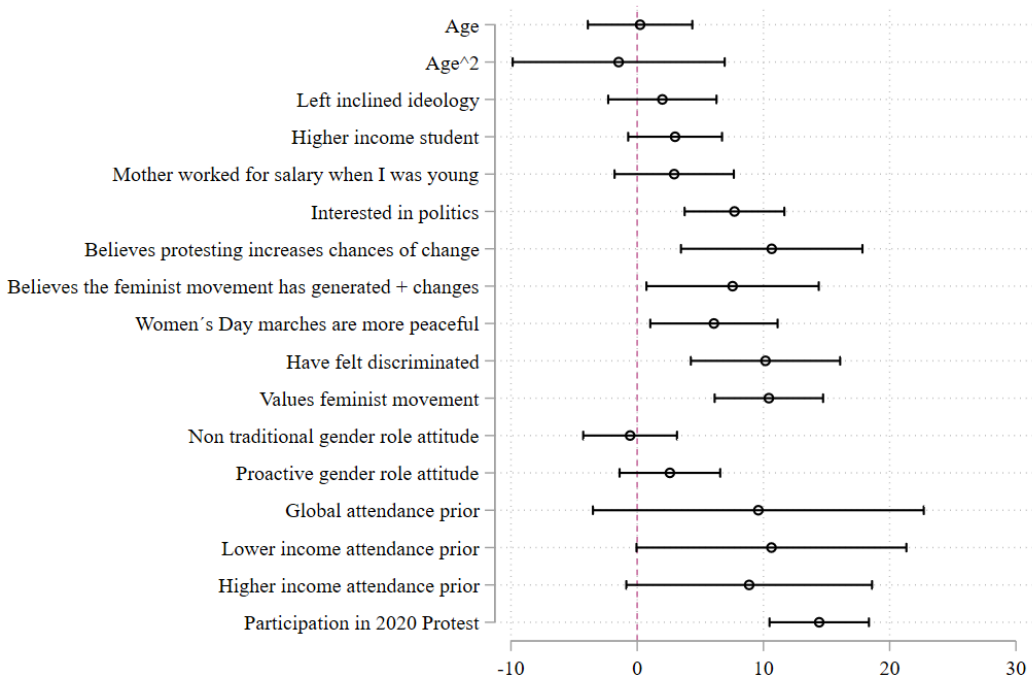
We find that important drivers of protest participation in this sample are: past participation in protests, past experiences of discrimination, appreciation of the feminist movement, the belief that protesting increases chances of change, interest in politics, attendance belief priors, and also being of a higher income. Another significant driver is the belief that marches are more peaceful, which is in line with associated literature that mention violence and repression in manifestations as important determinants of participation (DeNardo, 2014). Furthermore, income level (defined as belonging to the lower or higher income group) is seen to be a more significant and powerful predictor of participation than left-inclined political ideology: while an increase in income group from lower to higher is seen to increase the probability of participation by 6.3 percentage points (significant to a 5% level), left inclined ideology has a non-significant coefficient of 1.99 (this would mean, if significant, that a change in ideology from right to left[11] would increase probability of participation in 1.99 pp).

Past participation is certainly a very important motivator: participation in the previous year's protest increases the probability of participation by 14 to 18 percentage points, significant to a 1% level. This validates findings of related literature such as Chenoweth et al. (2022) and Bursztyn et al. (2021) where evidence of persistence in participation is found. In fact, this result is even more direct than those cited, as here we find that past participation increases the probability of attending the same protest, while in Chenoweth et al. (2022), they find that participation in some protests increases the chances of participating in other protests and Bursztyn et al. (2021) finds persistent participation in Hong Kong protests if a sufficient number of people from an individual's network is also incentivized to participate. The persistent nature of Women's Day marches and it's popularity allows us to evaluate this particular persistence in protests, and we find that it is quite an important effect to consider.

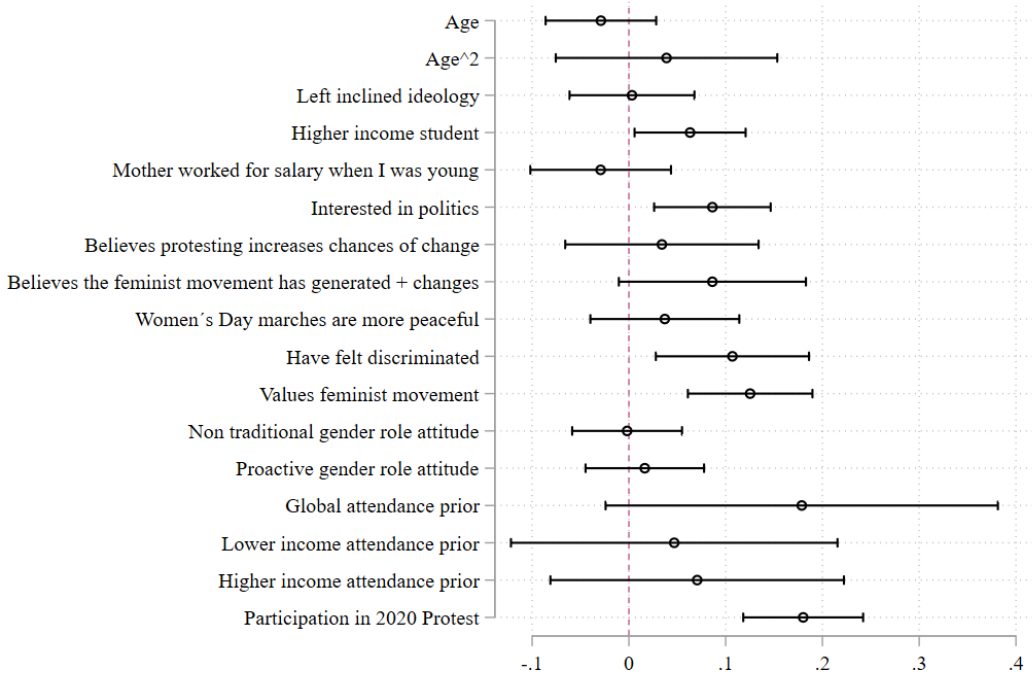
Importantly, group attendance priors also predict individual participation, but only at the 10% level and for the probability of protest (Panel A of Figure A-1), confirming, therefore, the existence of social drivers and strategic participation in the decision to protest. Although

[11] In the survey participants identified themselves in an ideological scale from 1 to 10 where 1 was left and 10 was right. To form two groups, and hence the binary dummy used in this regression, left was defined as those who identified from 1 to 4 on the scale, and the rest as right.

Figure A-1: Participation Motives



(a) Probability of Participating



(b) Effective Participation

Plots show point estimates and 95% confidence intervals of linear regressions using protest participation (measured as intention or effective participation) as a dependent variable. Both universities are considered within the studied sample.

Table A-1: **Why protest?**

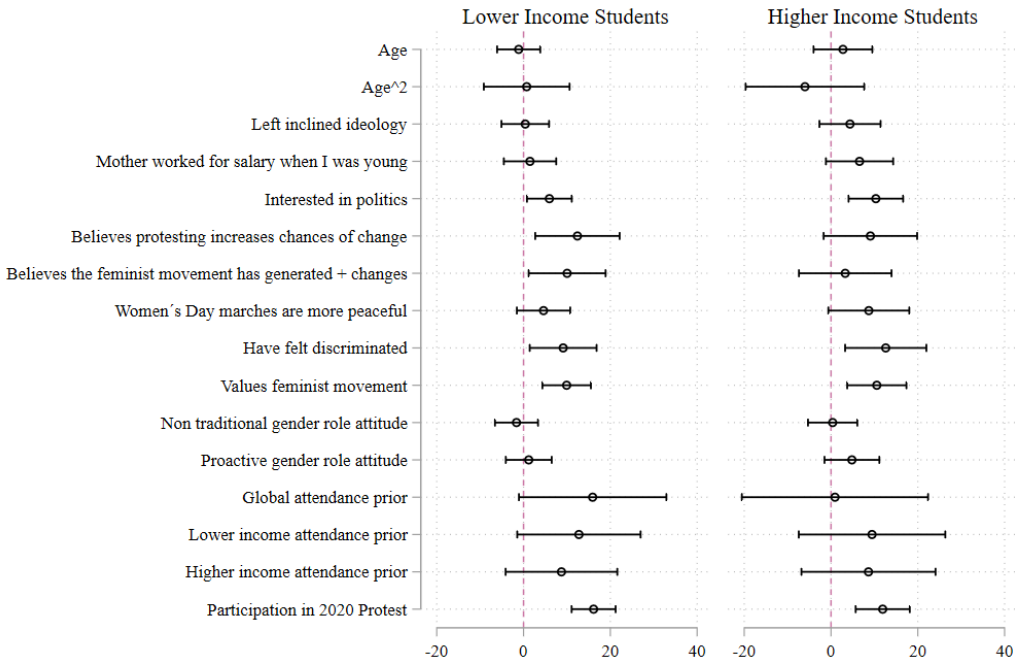
	Probability of Attending (Wave 2) (1)	Effective Participation (Wave 3) (2)
Age	0.229	-0.029
	(2.108)	(0.029)
Age^2	-1.468	0.039
	(4.280)	(0.058)
Left inclined ideology	1.995	0.003
	(2.186)	(0.033)
Higher income student	3.002	0.063**
	(1.895)	(0.029)
Mother worked for salary when I was young	2.930	-0.029
	(2.407)	(0.037)
Interested in politics	7.706***	0.086***
	(2.013)	(0.031)
Believes protesting increases chances of change	10.655***	0.034
	(3.663)	(0.051)
Believes the feminist movement has generated + changes	7.557**	0.086*
	(3.479)	(0.049)
Women´s Day marches are more peaceful	6.078**	0.037
	(2.571)	(0.039)
Have felt discriminated	10.162***	0.107***
	(3.014)	(0.040)
Values feminist movement	10.430***	0.125***
	(2.189)	(0.033)
Non traditional gender role attitude	-0.560	-0.002
	(1.893)	(0.029)
Proactive gender role attitude	2.588	0.016
	(2.032)	(0.031)
Global attendance prior	9.597	0.179*
	(6.679)	(0.103)
Lower income attendance prior	10.635*	0.047
	(5.448)	(0.086)
Higher income attendance prior	8.870*	0.071
	(4.960)	(0.077)
Participation in 2020 Protest	14.418***	0.180***
	(2.007)	(0.032)
Mean of Dep. Var.	59.57	0.45
St. Dev. of Dep. Var.	34.94	0.50
Observations	1,125	1,120

* p<0.10; ** p<0.05; *** p<0.01. A linear regression is estimated using protest attendance (measured as probability of attending or effective participation) as a dependent variable. Both universities are considered within the studied sample. Robust standard errors are reported in parenthesis.

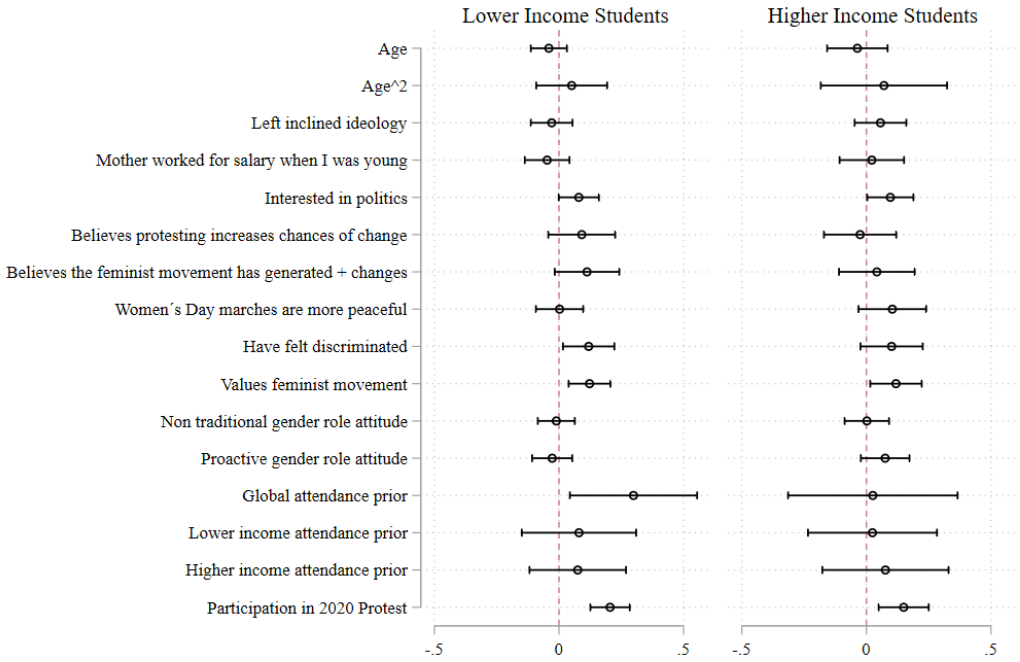
weak in significance, these coefficients have an important magnitude: depending on the attendance prior in question (global, higher income, or lower income attendance), an increase of 1 percentage point (pp) in these beliefs can increase the probability of participation in 8 to 17 pp.

When analyzed by income group (Figure A-2), one can observe certain heterogeneities in drivers of protest participation. Having a mother with a working salary when young seems to be a determinant for higher income but not for lower income students. Interest in politics is significant for both income levels, but has double the magnitude for higher income students (10.33 versus 5.95 percentage points). Interestingly, the belief that protests increase chances of change is only significant for lower income students and has an important magnitude of 12.41 percentage points. Global attendance priors are significant for the lower income group and have a similar effect as past participation: an increase in global attendance beliefs for lower income students increases the probability of participation by 15.93 percentage points. Past participation is equally significant for both income levels, but has a higher magnitude for lower income students (16.1 versus 11.91 percentage points). Finally, it is important to note that both income groups' valuation (or appreciation) of the feminist movement drive participation similarly and at the same significance.

Figure A-2: Participation Motives by Income Group



(a) Probability of Participating



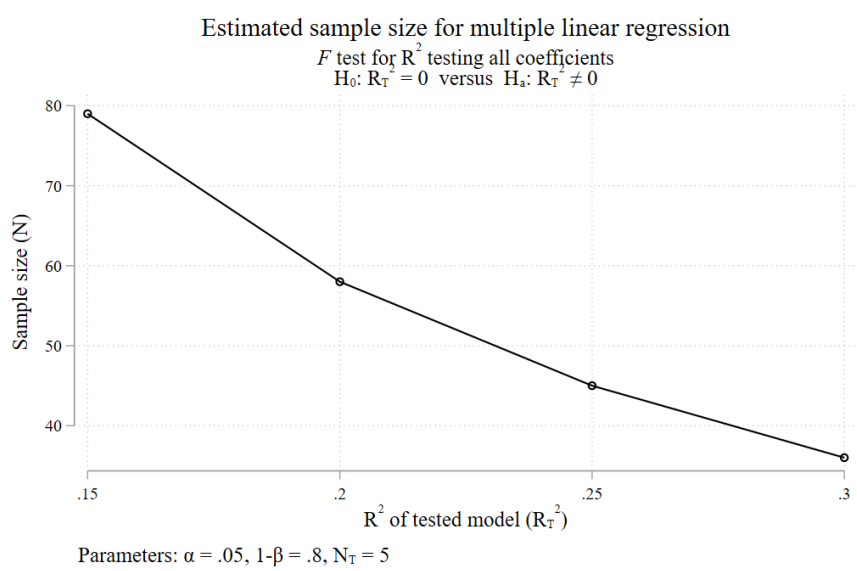
(b) Effective Participation

Plots show point estimates and 95% confidence intervals of linear regressions using protest attendance (measured as probability of attending or effective participation) as a dependent variable. Two set of estimates are shown for each dependent variable: estimates for lower and higher income group students. Both universities are considered within the studied sample.

B Power Analysis

For these analyses we use the most intensive estimation in samples and covariates as a parameter; the first stage estimation which has 5 covariates (one for each treatment) and two sub-samples (estimations must be run separately for higher income and lower income). When setting power ideally to 80% and assuming a significance criteria of 5%, we find that depending on the R^2 parameter being used[12], required sample sizes vary between 36 and 79 observations per sub-sample (income group). These sample sizes are more than achieved: for that same first stage estimation final sub-samples varied between 200 - 600 observations. These sub-samples consider both universities and all treatments, as estimations pool these groups together (while controlling for them).

Figure B-1: Power Estimations: Sample size required for determined R^2 levels)

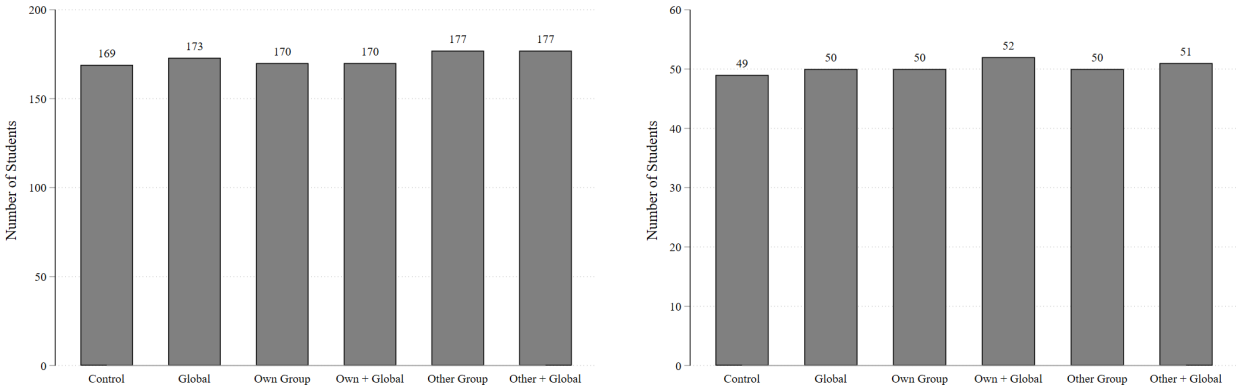


Power estimations were run using the F test of significance of R^2 as we are interested in the significance of all covariates. Power = 0.8, and α=0.05.

[12] As we are interested in the significance of all of the treatment predictors together, we will use the F test of significance of R^2 which is recommended for these type of analyses.

C Additional Figures

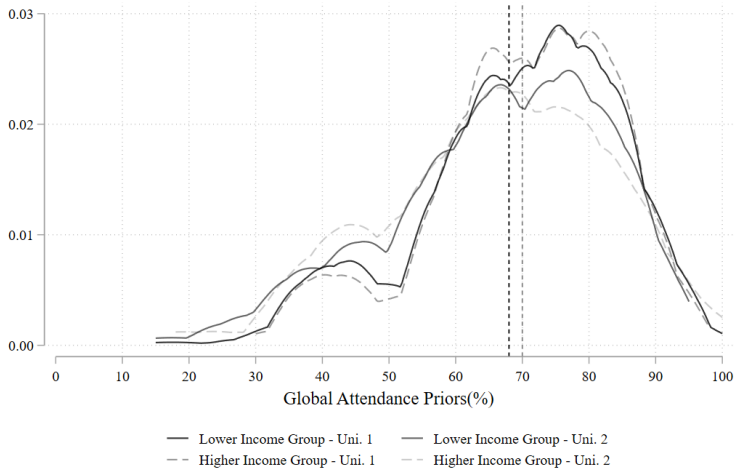
Figure C-1: Treatment Distribution



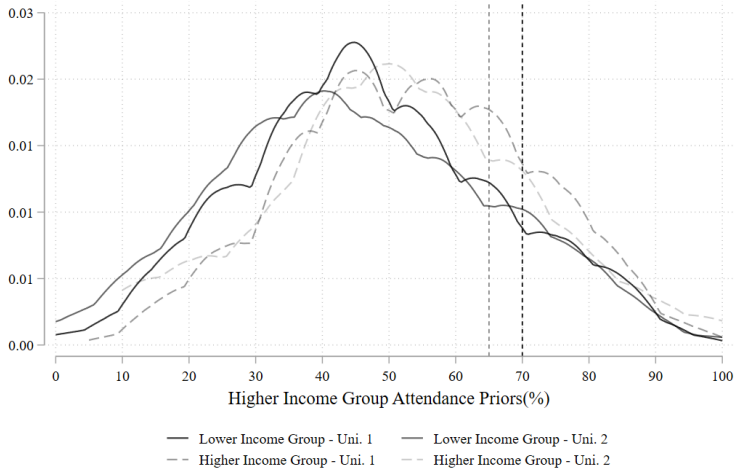
(a) University 1 (b) University 2

Bar graphs portray the distribution of subjects in both universities along treatments. The survey server was instructed to assign treatments in a balanced manner, as subjects entered the questionnaire.

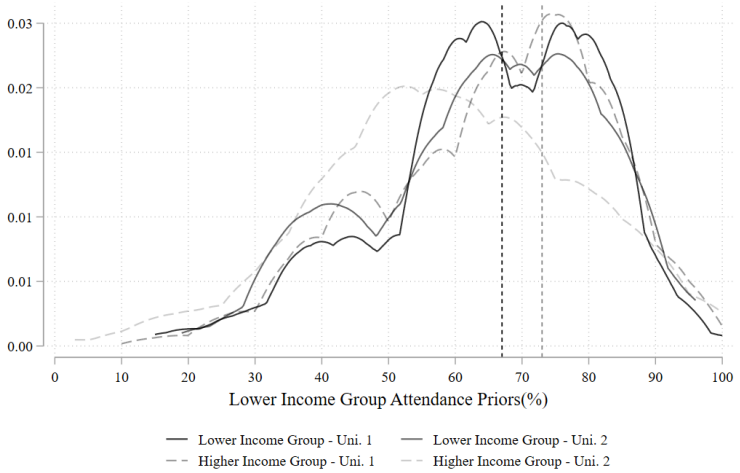
Figure C-2: Distribution of Priors



(a) Global Participation



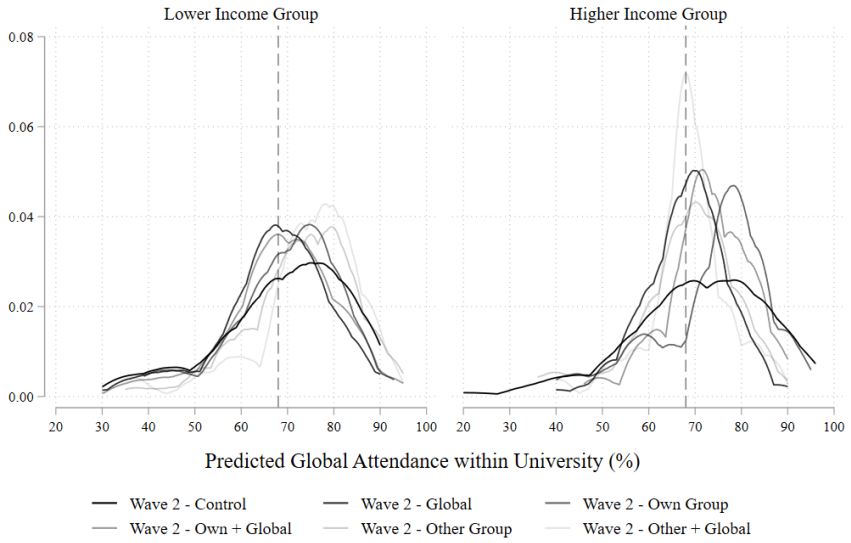
(b) Higher-Income Group Participation



(c) Lower-Income Group Participation

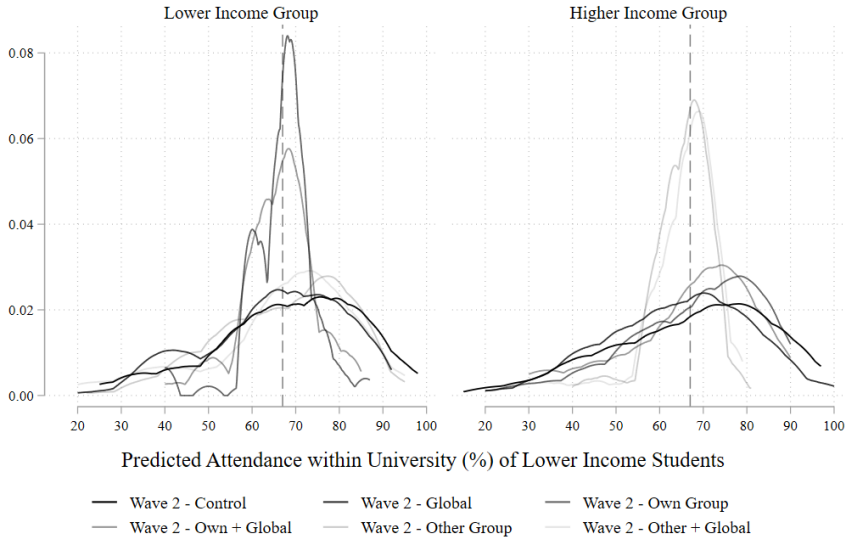
Kernel density estimates are displayed for the distribution of global, higher and lower income participation priors of the 4 different samples chosen to study: lower and higher income groups of both universities. Vertical dashed lines show treatment truths for both universities of the chosen prior.

Figure C-3: Posteriors about Global Participation



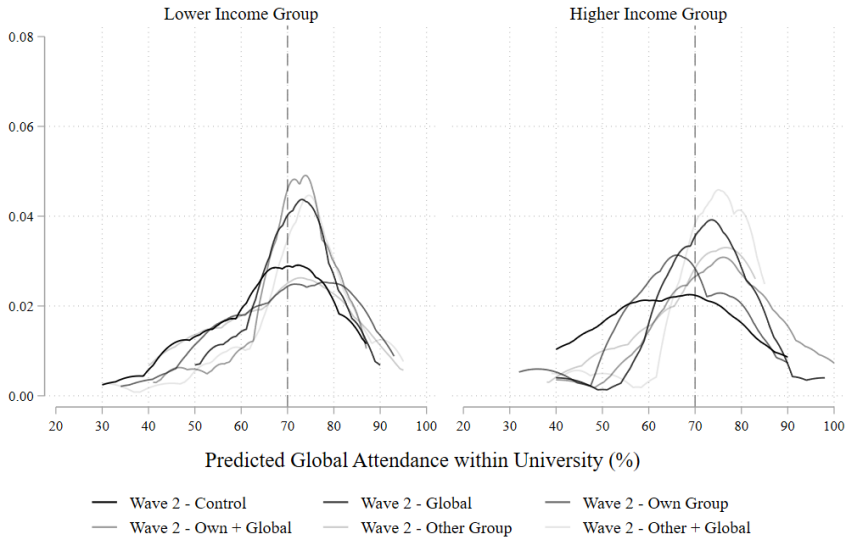
Kernel density estimates are displayed for global attendance beliefs for both income groups (shown in different panels) and all treatments (different colored lines). Sample considers only University 1 and beliefs are posteriors, i.e. post receiving the information treatment. The vertical dashed line is the treatment truth delivered.

Figure C-4: Posteriors about Participation of Lower Income Group



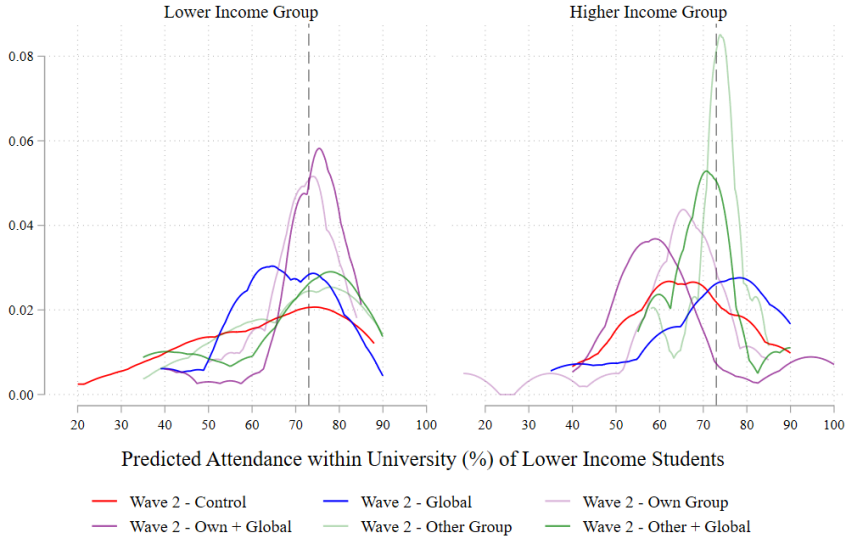
Kernel density estimates are displayed for lower income attendance beliefs for both income groups (shown in different panels) and all treatments (different colored lines). Sample considers only University 1 and beliefs are posteriors, i.e. post receiving the information treatment. Vertical dashed line is the treatment truth delivered.

Figure C-5: Global Attendance Posteriors



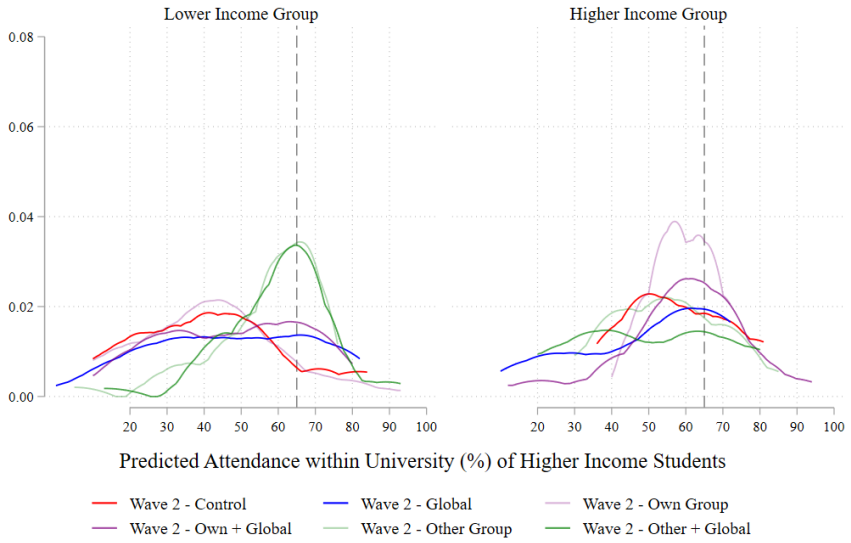
Kernel density estimates are displayed for global attendance beliefs for both income groups (shown in different panels) and all treatments (different colored lines). Sample considers only University 2 and beliefs are posteriors, i.e. post receiving the information treatment. Vertical dashed line is the treatment truth delivered.

Figure C-6: Lower Inc. Group Attendance Posteriors



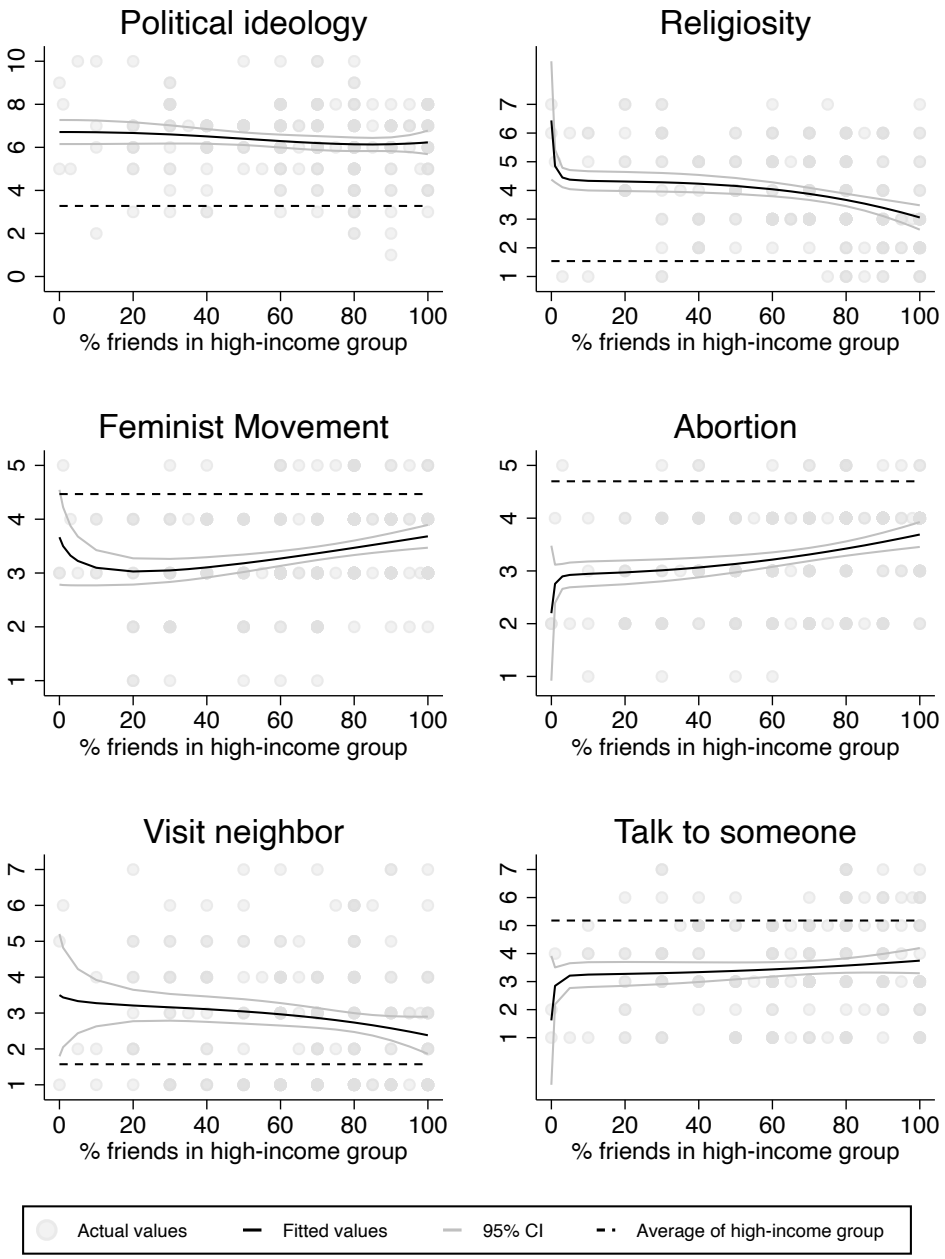
Kernel density estimates are displayed for lower income attendance beliefs for both income groups (shown in different panels) and all treatments (different colored lines). Sample considers only University 2 and beliefs are posteriors, i.e. post receiving the information treatment. Vertical dashed line is the treatment truth delivered.

Figure C-7: Higher Inc. Group Attendance Posteriors



Kernel density estimates are displayed for higher income attendance beliefs for both income groups (shown in different panels) and all treatments (different colored lines). Sample considers only University 2 and beliefs are posteriors, i.e. post receiving the information treatment. Vertical dashed line is the treatment truth delivered.

Figure C-8: Students' Interactions and In-Group Misperceptions: Higher Income Group



Notes: In each panel, the horizontal axis plots the percentage of high-income friends students declare to have, and the vertical axis corresponds to each dimension studied. The dashed line shows the actual average response of the high-income group, and the continuous curve shows the fitted value for the high-income students' perceptions of their own group's responses.

D Additional Tables

Table D-1: **Historic Participation in Women's Day March in Santiago, Chile**

Year	Estimated Number of Participants	Hurt or Detained
2019	180.000	36
2020	1.000.000	35
2021	300.000	15
2022	300.000	-
2023	-	-

These statistics were created by averaging the number of participants detailed in different articles, newspapers, media, etc. found online.

Table D-2: **Wave 1 Filters**

	Participants
Total	2944
Under age (-18)	129
Other University	33
Don't accept informed consent	7
Male or Prefer not to answer	588
No name	121
Duplicates	13
Final Participants	2053

Table D-3: **Analisis of Attrition between Waves 1 and 2**

	Response Wave 2
Intention to Participate	-0.0135
	[0.0124]
Appreciation of Feminist Movement	0.0007
	[0.0148]
Would you characterize Women´s Day Protests as violent?	-0.0045
	[0.0155]
Political ideology (1=Left, 10=Right)	-0.0090
	[0.0066]
Interest in politics	0.0045
	[0.0120]
Monthly Income Range	0.0006
	[0.0064]
Have you ever felt discriminated?	-0.0405
	[0.0313]
Participation in 2020 March	0.0382
	[0.0232]
Belief of police repression in this upcoming protest	-0.0004
	[0.0004]
Age	0.0041
	[0.0034]
Year of Admission to Uni.	0.0002*
	[0.0001]
Constant	0.4270*
	[0.2467]
Observations	1510

* p<0.10; ** p<0.05; *** p<0.01. A linear regression was estimated using a binary variable indicating if subjects responded the second wave of the survey as a dependent variable, in order to determine if there were any underlying determinants of attrition. Robust standard errors are reported in parenthesis.

Table D-4: **Characterization of Experimental Sample**

	Mean	St. Dev.	Min.	Max.	Obs.
Panel A: Outcome Variables					
Intention to participate - Wave 1	2.96	0.97	1.00	4.00	1310
Intention to participate - Wave 2	2.92	1.05	1.00	4.00	1314
Recoded Intention to participate - Wave 2	0.65	0.48	0.00	1.00	1338
Probability of Participation - Wave 1	60.08	31.88	0.00	100.00	1337
Probability of Participation - Wave 2	58.36	34.90	0.00	100.00	1336
Final participation	0.44	0.50	0.00	1.00	1323
Global Attendance Prior	69.05	14.90	15.00	100.00	1338
Global attendance Posterior	70.29	12.14	20.00	100.00	1338
Higher Income Attendance Prior	49.34	19.20	0.00	100.00	1336
Higher Income attendance Posterior	54.02	18.25	0.00	100.00	1338
Lower Income Attendance Prior	64.88	16.77	3.00	100.00	1338
Lower Income attendance Posterior	66.87	13.95	15.00	100.00	1338
Panel B: Other Characteristics					
Ideological Position(1=Left, 10=Right)	3.05	1.95	0.00	10.00	1337
Level of interest in Politics	2.35	1.00	0.00	4.00	1329
Frequency of conversation about Feminist topics	3.45	0.77	1.00	5.00	1337
Appreciation of Feminist Movement	4.46	0.83	1.00	5.00	1338
Movilization leads to more change	4.38	0.78	1.00	5.00	1324
Violence is justified in some cases	2.70	1.20	1.00	5.00	1301
Women´s movilization has led to positive changes nationally	4.38	0.73	1.00	5.00	1328
Level of violence in Women´s Day Protests (1=Peaceful, 5=Violent)	1.80	0.76	1.00	5.00	1337
Probability of repression in Women´s Day Protests	50.23	25.17	0.00	100.00	1337
Social Class	2.78	1.19	1.00	5.00	1305
Monthly Household Income Range	2.65	1.65	1.00	6.00	1338
Age	21.10	3.21	18.00	45.00	1336
Have felt discriminated	0.84	0.37	0.00	1.00	1204
Mother worked for salary when I was young	0.82	0.38	0.00	1.00	1276
Average participation in marches 2019-2022	0.39	0.33	0.00	1.00	1329
Non Traditional Gender Role Attitude	4.11	0.50	2.00	5.00	1338
Proactive Gender Role Attitude	4.26	0.56	1.50	5.00	1338
Believes women are discriminated in life	4.54	0.58	1.00	5.00	1338

The studied sample considers both universities and both income groups. The variable *Recoded Intention to Participate* is the result of transforming *Intention to Participate - Wave 2* into a dummy. This is done by setting *Most likely to not attend* and *Will not attend* equal to 0, and *Most likely to attend* and *Will attend* equal to 1.

Table D-5: **Characterization of Treatment Groups**

	Treatment						
	Control (1)	Global (2)	Own (3)	Own + Global (4)	Other (5)	Other + Global (6)	F-Test (7)
Ideological Position(1=Left, 10=Right)	3.28	3.05	2.99	3.00	3.04	2.96	0.75
Level of interest in Politics	2.33	2.31	2.37	2.39	2.34	2.38	0.21
Frequency of conversation about Feminist topics	3.36	3.47	3.45	3.50	3.45	3.47	0.77
Appreciation of Feminist Movement	4.43	4.49	4.44	4.38	4.56*	4.46	1.25
Movilization leads to more change	4.34	4.43	4.32	4.27	4.54***	4.38	3.37***
Violence is justified in some cases	2.69	2.77	2.59	2.71	2.74	2.70	0.58
Women´s movilization has led to positive changes nationally	4.36	4.43	4.39	4.27	4.46	4.38	1.8
Level of violence in Women´s Day Protests (1=Peaceful, 5=Violent)	1.75	1.79	1.80	1.84	1.80	1.82	0.36
Probability of repression in Women´s	50.27	52.80	46.34	51.46	50.33	50.13	1.62
Social Class	2.73	2.74	2.88	2.80	2.77	2.79	0.43
Monthly Household Income Range	2.61	2.60	2.74	2.62	2.67	2.67	0.22
Age	21.49	20.94*	21.15	20.96*	21.10	20.97	0.93
Have felt discriminated	0.83	0.83	0.85	0.83	0.82	0.87	0.61
Mother worked for salary when I was young	0.81	0.78	0.89**	0.85	0.82	0.79	2.43**
Average participation in marches 2019-2022	0.39	0.40	0.41	0.36	0.39	0.41	0.61
Non Traditional Gender Role Attitude	4.10	4.12	4.08	4.08	4.14	4.13	0.48
Proactive Gender Role Attitude	4.21	4.28	4.24	4.27	4.26	4.28	0.58
Believes women are discriminated in life	4.54	4.54	4.54	4.56	4.56	4.48	0.58

* p<0.10; ** p<0.05; *** p<0.01. Table shows the mean of several studied characteristics by treatment group. Stars in columns (1)-(7) represent p-values of t-tests between treatments and the control group. In addition, column (7) reports the F-test of a one-way analysis of variance (ANOVA), where larger values indicate greater differences between group means. The analyzed sample considers both universities.

Table D-6: **Characterization of Treatment Groups: University 1**

	Treatment Group												F-Test
	Control		Global		Own		Own + Global		Other		Other + Global		
	Mean (1)	Sample (2)	Mean (3)	Sample (4)	Mean (5)	Sample (6)	Mean (7)	Sample (8)	Mean (9)	Sample (10)	Mean (11)	Sample (12)	(13)
Panel A: Lower Income Group													
Ideological Position(1=Left, 10=Right)	2.96	99	2.95	101	2.99	97	2.87	98	2.92	101	2.83	101	0.10
Level of interest in Politics	2.39	99	2.21	98	2.26	96	2.43	97	2.28	101	2.40	100	0.87
Frequency of conversation about Feminist topics	3.40	99	3.48	101	3.36	97	3.46	98	3.35	101	3.46	101	0.48
Appreciation of Feminist Movement	4.49	99	4.46	101	4.48	97	4.30	98	4.51	101	4.43	101	0.86
Movilization leads to more change	4.37	98	4.45	101	4.34	94	4.32	97	4.52	100	4.43	100	0.98
Violence is justified in some cases	2.80	94	2.94	100	2.63	92	2.92	96	2.79	96	2.85	100	0.82
Women´s movilization has led to positive changes nationally	4.45	98	4.44	99	4.39	96	4.28	96	4.48	100	4.36	101	0.95
Level of violence in Women´s Day Protests (1=Peaceful, 5=Violent)	1.82	99	1.87	101	1.90	97	1.97	97	1.86	101	1.83	101	0.46
Probability of repression in Women´s Day Protests	53.05	99	52.78	101	45.35**	97	55.29	98	50.59	101	49.17	101	2.10*
Social Class	1.94	97	1.98	97	2.16*	94	2.10	98	2.02	96	2.04	98	0.90
Monthly Household Income Range	1.48	99	1.47	101	1.58	97	1.47	98	1.44	101	1.50	101	0.93
Age	21.22	99	20.50	101	20.61	97	20.80	97	20.66	101	20.12**	101	0.31
Have felt discriminated	0.80	92	0.81	89	0.88	89	0.84	87	0.83	92	0.90*	86	0.90
Mother worked for salary when I was young	0.74	90	0.73	94	0.93***	90	0.77	92	0.74	98	0.79	95	3.07**
Average participation in marches 2019-2022	0.37	97	0.38	100	0.36	96	0.34	98	0.34	101	0.39	99	1.66
Non Traditional Gender Role Attitude	4.08	99	4.10	101	4.04	97	3.96*	98	4.11	101	4.14	101	0.39
Proactive Gender Role Attitude	4.25	99	4.26	101	4.24	97	4.27	98	4.32	101	4.31	101	0.85
Believes women are discriminated in life	4.58	99	4.53	101	4.52	97	4.48	98	4.56	101	4.49	101	0.40
Panel A: Higher Income Group													
Ideological Position(1=Left, 10=Right)	3.57	70	3.29	72	2.90*	73	3.24	72	3.25	76	3.05	76	0.87
Level of interest in Politics	2.36	70	2.28	72	2.48	73	2.43	72	2.32	74	2.46	76	0.45
Frequency of conversation about Feminist topics	3.30	70	3.46	72	3.51*	73	3.50	72	3.53*	76	3.43	76	0.84
Appreciation of Feminist Movement	4.24	70	4.46	72	4.34	73	4.40	72	4.47	76	4.53**	76	1.01
Movilization leads to more change	4.25	68	4.35	71	4.27	73	4.24	72	4.49	76	4.32	76	0.84
Violence is justified in some cases	2.55	67	2.46	70	2.46	71	2.42	71	2.58	74	2.48	73	0.19
Women´s movilization has led to positive changes nationally	4.26	70	4.38	71	4.38	72	4.23	71	4.46	76	4.45	76	1.31
Level of violence in Women´s Day Protests (1=Peaceful, 5=Violent)	1.70	70	1.67	72	1.82	73	1.76	72	1.84	76	1.83	76	0.71
Probability of repression in Women´s Day Protests	44.69	70	46.90	72	45.75	72	46.26	72	49.62	76	49.24	76	0.43
Social Class	3.91	68	3.87	70	3.96	71	3.87	69	3.77	75	3.91	74	0.53
Monthly Household Income Range	4.36	70	4.40	72	4.52	73	4.39	72	4.49	76	4.41	76	0.19
Age	21.39	70	20.94	72	21.14	73	20.96	72	21.46	76	21.07	76	0.29
Have felt discriminated	0.79	62	0.79	63	0.86	69	0.86	64	0.79	66	0.89	66	0.94
Mother worked for salary when I was young	0.90	69	0.85	68	0.87	71	0.99**	69	0.84	76	0.77**	74	3.22**
Average participation in marches 2019-2022	0.42	70	0.41	72	0.41	73	0.36	72	0.38	76	0.40	76	0.36
Non Traditional Gender Role Attitude	4.14	70	4.29*	72	4.19	73	4.19	72	4.22	76	4.19	76	0.76
Proactive Gender Role Attitude	4.14	70	4.28	72	4.15	73	4.29	72	4.22	76	4.32*	76	1.14
Believes women are discriminated in life	4.46	70	4.45	72	4.47	73	4.60	72	4.49	76	4.51	76	0.68

* p<0.10; ** p<0.05; *** p<0.01. Table shows the mean of several studied characteristics by treatment group. Stars in columns (1)-(12) represent p-values of t-tests between treatments

Table D-7: **Characterization of Treatment Groups: University 2**

	Treatment Group												F-Test
	Control		Global		Own		Own + Global		Other		Other + Global		
	Mean (1)	Sample (2)	Mean (3)	Sample (4)	Mean (5)	Sample (6)	Mean (7)	Sample (8)	Mean (9)	Sample (10)	Mean (11)	Sample (12)	(13)
Panel A: Lower Income Group													
Ideological Position(1=Left, 10=Right)	3.78	32	2.67**	33	2.91*	32	2.88*	34	2.76**	33	2.94*	33	1.55
Level of interest in Politics	1.91	33	2.55***	33	2.41**	32	2.38**	34	2.79***	33	2.09	32	3.32**
Frequency of conversation about Feminist topics	3.24	33	3.58*	33	3.59*	32	3.68**	34	3.61**	33	3.55*	33	1.28
Appreciation of Feminist Movement	4.48	33	4.64	33	4.50	32	4.74	34	4.85**	33	4.55	33	1.51
Movilization leads to more change	4.42	33	4.61	33	4.42	31	4.41	34	4.82***	33	4.39	33	2.23*
Violence is justified in some cases	2.56	32	2.91	33	2.91	32	2.79	34	2.84	32	2.72	32	0.36
Women´s movilization has led to positive changes nationally	4.27	33	4.55	33	4.41	32	4.44	34	4.52	33	4.27	33	0.92
Level of violence in Women´s Day Protests (1=Peaceful, 5=Violent)	1.70	33	1.91	33	1.56	32	1.65	34	1.58	33	1.79	33	1.15
Probability of repression in Women´s Day Protests	53.91	33	64.91	33	53.06	32	56.24	34	51.67	33	58.76	33	1.11
Social Class	2.18	33	2.00	32	2.10	31	2.12	34	2.03	33	1.91	33	0.82
Monthly Household Income Range	1.52	33	1.42	33	1.50	32	1.47	34	1.39	33	1.42	33	0.30
Age	22.15	33	21.85	33	22.09	32	20.91***	34	21.27*	33	22.33	33	1.64
Have felt discriminated	0.90	31	0.97	31	0.79	29	0.90	30	0.90	31	0.83	30	1.10
Mother worked for salary when I was young	0.75	32	0.77	31	0.83	29	0.76	33	0.94**	32	0.81	31	1
Average participation in marches 2019-2022	0.36	33	0.41	32	0.52*	32	0.46	33	0.55**	33	0.42	33	1.57
Non Traditional Gender Role Attitude	4.11	33	3.78**	33	3.93	32	4.17	34	4.09	33	3.95	33	2.64**
Proactive Gender Role Attitude	4.18	33	4.39*	33	4.37	32	4.26	34	4.32	33	4.22	33	0.77
Believes women are discriminated in life	4.49	33	4.75**	33	4.70*	32	4.65	34	4.69*	33	4.39	33	2.36**
Panel A: Higher Income Group													
Ideological Position(1=Left, 10=Right)	2.94	16	3.35	17	3.44	18	3.06	18	3.41	17	3.33	18	0.21
Level of interest in Politics	2.69	16	2.53	17	2.44	18	2.00*	18	1.94**	17	2.39	18	1.26
Frequency of conversation about Feminist topics	3.63	16	3.29	17	3.39	18	3.33	18	3.41	17	3.53	17	0.48
Appreciation of Feminist Movement	4.69	16	4.59	17	4.50	18	4.11*	18	4.71	17	4.28	18	1.49
Movilization leads to more change	4.44	16	4.35	17	4.28	18	3.89**	18	4.40	15	4.41	17	1.46
Violence is justified in some cases	2.94	16	2.82	17	2.28**	18	2.59	17	2.88	17	2.76	17	0.74
Women´s movilization has led to positive changes nationally	4.44	16	4.35	17	4.39	18	4.11	18	4.29	17	4.39	18	0.47
Level of violence in Women´s Day Protests (1=Peaceful, 5=Violent)	1.63	16	1.65	17	1.56	18	1.83	18	1.65	17	1.72	18	0.28
Probability of repression in Women´s Day Protests	49.94	16	54.41	17	42.11	18	42.44	18	49.41	17	43.44	18	0.68
Social Class	3.73	15	3.76	17	3.72	18	3.82	17	4.06	17	3.89	18	0.56
Monthly Household Income Range	4.13	16	4.00	17	3.94	18	3.94	18	4.29	17	4.17	18	0.27
Age	22.25	16	21.71	17	22.44	18	21.94	17	21.76	17	22.89	18	0.70
Have felt discriminated	1.00	15	0.81*	16	0.82*	17	0.56***	16	0.75**	16	0.76**	17	1.89
Mother worked for salary when I was young	0.88	16	0.71	17	0.83	18	0.88	17	0.94	17	0.88	17	0.85
Average participation in marches 2019-2022	0.49	16	0.47	17	0.49	18	0.31	17	0.41	17	0.47	18	0.87
Non Traditional Gender Role Attitude	3.98	16	4.14	17	4.12	18	4.20	18	3.98	17	4.16	18	0.74
Proactive Gender Role Attitude	4.33	16	4.25	17	4.40	18	4.23	18	3.95**	17	4.09	18	1.67
Believes women are discriminated in life	4.72	16	4.59	17	4.64	18	4.57	18	4.66	17	4.44	18	0.52

* p<0.10; ** p<0.05; *** p<0.01. Table shows the mean of several studied characteristics by treatment group. Stars in columns (1)-(12) represent p-values of t-tests between treatments and the

Table D-8: **First Stage:**
Treatment Effects on Posterior Beliefs (Including interactions of treatments)

	Attendance Posteriors					
	Global		Lower Income		Higher Income	
	Prior Below Truth (1)	Prior Above Truth (2)	Prior Below Truth (3)	Prior Above Truth (4)	Prior Below Truth (5)	Prior Above Truth (6)
Panel A: Lower Income Students						
Global Treatment	2.781**	-1.844**				
	(1.334)	(0.757)				
Own Treatment			8.898***	-5.789***		
			(1.458)	(1.298)		
Other Treatment					19.773***	-2.503
					(1.402)	(3.165)
Observations	317	467	419	365	651	133
Panel B: Higher Income Students						
Global Treatment	2.465*	-2.716***				
	(1.345)	(0.862)				
Other Treatment			8.901***	-6.486***		
			(1.955)	(1.388)		
Own Treatment					13.543***	-0.695
					(1.578)	(2.940)
Observations	215	326	276	265	412	129

* p<0.10; ** p<0.05; *** p<0.01 Regressions include the following controls: university dummy, value of feminist movement, protest is characterized as peaceful/violent, political ideology, interest in politics, past participation in Women's Day marches, commune of residence and attendance priors (global and relative). Robust standard errors are reported in parenthesis. Both universities are included in this analyzed sample.

Table D-9: **Posterior Effects on Participation, above and below truth together**

	Probability to Attend (Wave 2)			Effective Participation (Wave 3)		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Lower Income Students						
Global Attendance Posteriors	0.187*			0.172		
	(0.109)			(0.169)		
Lower Income Attendance Posteriors		-0.051			-0.160	
		(0.087)			(0.134)	
Higher Income Attendance Posteriors			0.118*			0.106
			(0.061)			(0.097)
Mean of Dep. Var.	57.22	57.22	57.22	42.06	42.06	42.06
St. Dev. of Dep. Var.	34.66	34.66	34.66	49.40	49.40	49.40
Observations	783	783	783	775	775	775
Panel B: Higher Income Students						
Global Attendance Posteriors	-0.102			-0.105		
	(0.177)			(0.243)		
Lower Income Attendance Posteriors		-0.028			-0.028	
		(0.110)			(0.173)	
Higher Income Attendance Posteriors			0.142			0.016
			(0.091)			(0.144)
Mean of Dep. Var.	60.11	60.11	60.11	47.11	47.11	47.11
St. Dev. of Dep. Var.	35.42	35.42	35.42	49.96	49.96	49.96
Observations	540	540	540	537	537	537

* p<0.10; ** p<0.05; *** p<0.01 Regressions include the following controls: university dummy, value of feminist movement, protest is characterized as peaceful/violent, political ideology, interest in politics, past participation in Women's Day marches, commune of residence and attendance priors(global and relative). Robust standard errors are reported in parenthesis. Both universities are included in this analyzed sample.

Table D-10: **Two Stage Estimates: First Stage Results**

	Sample: Lower Income Students				Sample: Higher Income Students			
	Prior Below Truth (1)	Prior Above Truth (2)	Prior Below Truth (3)	Prior Above Truth (4)	Prior Below Truth (5)	Prior Above Truth (6)	Prior Below Truth (7)	Prior Above Truth (8)
Panel A: Global Attendance Posteriors								
Global Treat.	2.166	-1.800**	2.280	-1.926**	2.206	-2.678***	2.136	-2.704***
	(1.378)	(0.762)	(1.403)	(0.763)	(1.374)	(0.892)	(1.387)	(0.894)
Mean of Dep. Var.	61.67	76.16	61.63	76.14	62.27	75.32	62.26	75.34
St. Dev. of Dep. Var.	12.84	8.11	12.91	8.09	11.82	8.12	11.86	8.14
Observations	316	467	311	464	214	326	213	324
Panel B: Own Group Attendance Posteriors								
Own Treat.	6.990***	-5.058***	6.852***	-4.902***	13.863***	-1.985	14.026***	-1.922
	(1.283)	(1.135)	(1.303)	(1.129)	(1.382)	(2.608)	(1.380)	(2.752)
Mean of Dep. Var.	61.09	73.88	61.07	73.93	53.11	68.51	53.20	68.53
St. Dev. of Dep. Var.	14.02	10.62	14.07	10.58	16.27	12.19	16.34	12.28
Observations	418	365	413	362	411	129	410	127
Panel C: Other Group Attendance Posteriors								
Other Treat.	19.378***	1.230	19.372***	1.230	6.430***	-6.587***	6.330***	-6.698***
	(1.196)	(2.548)	(1.194)	(2.548)	(1.553)	(1.031)	(1.551)	(1.041)
Mean of Dep. Var.	48.60	68.80	48.81	68.80	60.19	72.85	60.31	72.84
St. Dev. of Dep. Var.	18.19	13.66	18.07	13.66	13.99	10.19	14.05	10.21
Observations	650	133	642	133	275	265	273	264

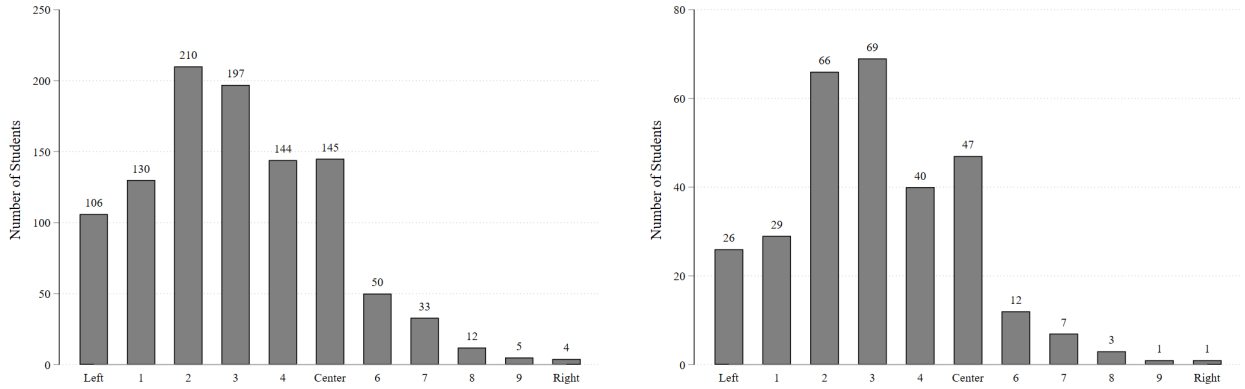
* p<0.10; ** p<0.05; *** p<0.01 This table reports the first stage of the principal two stage estimates reported in 7. The same regression is run in columns (1) & (3), (2) & (4), (5) & (7), and (6) & (8), but results vary minimally as samples will change depending on the final dependent variable used in second stage (either reported probability to attend or effective participation). Regressions include the following controls: university dummy, value of feminist movement, protest is characterized as peaceful/violent, political ideology, interest in politics, past participation in Women's Day marches, commune of residence and attendance priors(global and relative). Robust standard errors are reported in parenthesis. Both universities are included in this analyzed sample.

E Determination of Groups and Dimension of Heterogeneity

After the first wave of the survey and an initial characterization of the sample, the decision was taken to utilize income level as the dimension of heterogeneity to form groups. Both political ideology and income level were considered, but income level was finally chosen due to several reasons. The most important being that income level allowed us to separate the sample into two groups in a more absolute manner. If groups are determined using self-reported ideology in a scale from 0 to 10, as the one used in this survey, how do you define what is left, right and center? Someone who self-reports themselves as 3 and is therefore maybe assigned to the group left for example, might really feel more close to the center group. When receiving information treatments this could affect results due to misaligned own/other group identification. In order to avoid such issues, income level was chosen as it is not subject to such interpretation problems.

Another reason behind this decision was that the sample was notoriously biased towards the left (as seen in Figure E-1) in the political spectrum, and therefore it could have been a problem when creating groups if there was a lack of heterogeneity.

Figure E-1: Possible Dimension of Heterogeneity: Political ideology



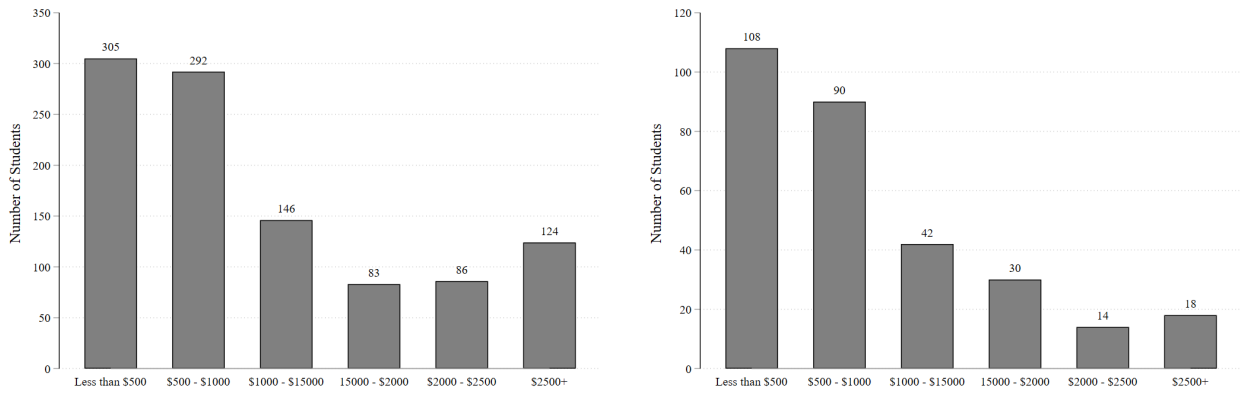
(a) University 1

(b) University 2

Several questions within the first survey attempt to gather information about income level: identification with a social class (low, medium, or high), housing commune, and monthly family income are all asked. But the last mentioned of these is chosen to create the groups for the intervention as it avoids the problems discussed above (interpretation and distribution).

Students were asked to report family monthly income levels within ranges of $ 500.000 CLP from $0 to +$2.500.000 CLP. The distribution of students within each university according to income level can be observed in Figure E-2.

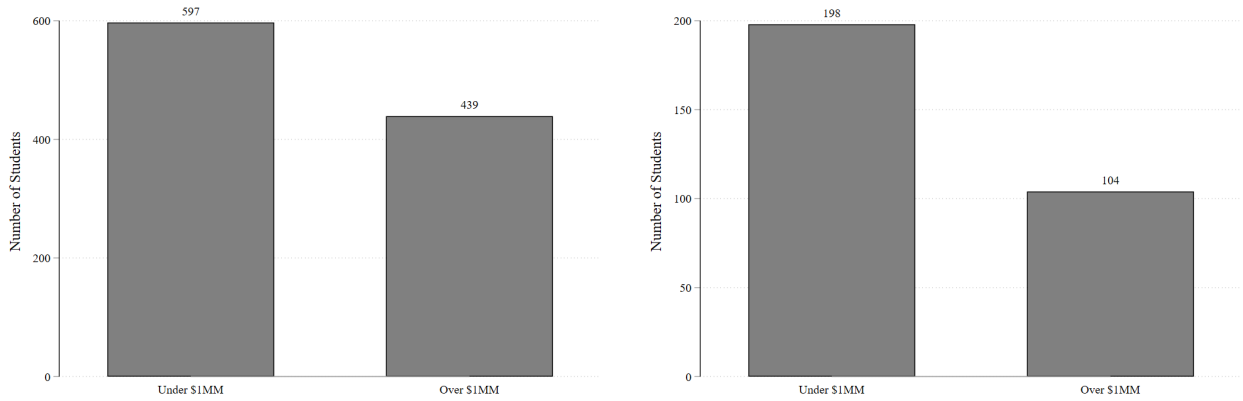
Figure E-2: Possible Dimension of Heterogeneity: Family Monthly Income



(a) University 1 (b) University 2

This income variable was utilized to divide students into two aprox. equal groups: below $1.000.000 CLP and above $1.000.000 CLP, as shown in E-3. Treatments in wave 2 will therefore consider subject's group and university in order to report estimated attendance.

Figure E-3: Groups formed according to Family Income Level



(a) University 1 (b) University 2

F Ethical Considerations

As it is standard in any field experiment that involves ethical considerations, we obtained IRB ethical approval to ensure that our experiment did not violate any ethical concerns that could be raised by the ethics committees involved. However, it is still essential to discuss the main ethical concerns that might arise with our intervention.

On the one hand, our experiment carries the inherent risk associated with involving participants in an experiment linked to protests: potentially exposing subjects to a violent event. First, it is important to highlight that the Women's March is widely recognized for its peaceful nature, and this implies that participation imposes very low risks on participants. Our survey data supports this: participants rate their perceptions of violence in the march at an average of 1.80 on a scale from 1 (Peaceful) to 5 (Violent). Second, the information that we gave students was carefully restricted to the sample of participants. In that sense, we did not obtain information from any external or biased sources, and participants were not deceived in any way.

On the other hand, our field experiment was targeted at students who identified as women. This required us to be particularly mindful of respecting the students' preferences and complying with the policies of the universities involved. To address this, we conducted a series of meetings with the authorities responsible for Gender Equity, both at the university and at the school level. They provided us with invaluable guidance in implementing the experiment, particularly in writing the survey, defining the pool of potential participants, and addressing potential risks.